

primary care veterinarian to inform them of the self-referral and to ask for patient records (pending owner release).

A review was made of the medical notes from both the primary care veterinarian and the chiropractor. It was apparent that the patient had not been seen by the primary care veterinarian since she had developed neurologic signs. The patient had been clinically normal at last veterinary examination. The chiropractor's records noted a mentally alert but ataxic patient, palpation of the spine revealed atlanto-occipital joint restriction in motion. No spinal adjustments were performed, but manual therapy techniques were applied to the muscles of the neck (Box 21.6, red flag box 1). After two visits and no improvement in gait or comfort, the recommendation for veterinary examination was made. It was noted in the chiropractor's medical record for the dog that it

is "customary for chiropractors to advise clients that three chiropractic visits should take place before drawing conclusions about how much change we will be able to make through adjusting however concerns about patient status raised a concern before this. It was becoming clear that imaging should be performed. An evaluation from a rehabilitation veterinarian will help to narrow the specific area to be imaged" (Box 21.6, red flag box 2).

The review of the notes produced a tentative differential diagnosis list. Top of the list was spinal cord compression. It was the rehabilitation veterinarian's opinion that cervical vertebral malformation was the primary differential diagnosis, but that other forms of spinal compression, congenital issues with central nervous system development, or infectious/inflammatory disease of the nervous system were possible.

Box 21.6 Red flag boxes for Case 6

- 1) The animal chiropractor stressed the point that she had not manipulated the patient's joints in the areas of concern. Although manipulations have been found to be safe and effective in the management of discomfort due to nerve compression (Lisi and Bhardwaj, 2004) and have been shown to safely provide symptomatic relief in the case of cervical disc herniation (Peterson *et al.*, 2013), it was clinically advisable to use caution in a case where hypomobility may be a protective response to hypermobility from instability, and/or to impingement on the spinal cord
- 2) The history of acute progression from lameness to paresis indicates spinal cord compression. A patient with these signs should be fully evaluated by a veterinarian as soon as possible. Advanced imaging of the spinal cord is advised. A young, large-breed dog with neurologic signs may have cervical vertebral spondylomyelopathy/instability (wobblers syndrome) or other congenital abnormality
- 3) A patient who has a (tentative) diagnosis of instability in the vertebral column may have intermittent exacerbation of spinal cord compression on making certain movements—in this case, full range neck movements should be avoided, especially rapid motions as they involve more force in muscle activation which in turn results in more compressive forces on the vertebral joints. Traction on walking surfaces needs to be adequate and the patient needs to be supported as needed
- 4) A patient may improve in strength and coordination after rehabilitation therapy but this does not mean the underlying disease is completely managed or at all resolved. Age-related progression of signs in patients with vertebral instability are due to degenerative changes, disc degeneration, and hypertrophy of ligaments and other soft tissues. These changes can increase compression of the spinal cord. Owners should be cautioned about progression of signs and regular monitoring of these cases is needed

As the patient's condition was stable, an examination at the rehabilitation clinic was performed as requested. Referral to a neurologist would then be discussed with the owner after the physical and functional examination findings were explained.

Rehabilitation veterinarian examination findings included grade 2/4 ataxia of the rear limbs with low ground clearance in swing phase and wide-based stance which became wider on turning. On turning, the patient swayed in a frontal plane but did not fall. Transitions between postures (sit, down, stand) revealed reduced eccentric control of motion with some tendency to fall into a sit or a down after initiation of movement. Sit position was with the forelimbs placed in abduction and the rear limbs extended to one side. Head position was forward with the neck in slight flexion. Stance posture was head-low and wide-based rear limbs. The patient could rise from a down or sitting position without assistance but relied on forward motion of the head, neck, and front limbs. Rear extension was partial until the patient moved forward several steps after a stand was achieved. Voluntary motion of the cervical spine using treat motivation revealed normal range of motion in lateral and ventral flexion of the neck but reluctance to extend the neck beyond the neutral (in line with thoracic spine) position. Palpation of the spine revealed restriction in motion of the C4–C5 intervertebral joints and the atlanto-occipital joints. Neck musculature was noted to be high in tone but no specific areas of pain were identified. Segmental spinal reflexes were within normal limits, and there was minor weakness in the withdrawal reflexes of the rear limbs. Tail motion was normal.

Radiographic findings (tipping at vertebral end plates, narrowing of spinal canal) were consistent with signs of cervical spondylomyelopathy, and advanced imaging (MRI) along with consultation with a neurologist was recommended by the rehabilitation veterinarian. The owner declined as she was not prepared to “put her animal through subsequent surgery” based on previous experi-

ences she had undergone with a dog who had had the same disorder.

A therapeutic/management plan was developed. The patient would undergo in-clinic and home exercises aimed at working on strength and coordination while preventing overload and excess motion in the cervical spinal cord as much as possible. A cervical collar (neck brace) was custom ordered from a veterinary supply company (Figure 21.8) and was used to prevent large motions of the cervical spine. The patient was changed to a diet with relatively low energy intake, correct calcium:phosphorus balance and adequate protein in order to prevent developmental orthopedic disease. Daily caloric intake was calculated by the rehabilitation technician who was responsible for nutritional counseling as well as home care advice and teaching home therapeutic exercises. The rehabilitation veterinarian explained the disorder to the owner at length and advised about the potential for progression and management of that progression. Every team member was also educated about progression, precautions, and activity contraindications (Box 21.6, red flag box 3).

The rehabilitation technician carried out in-clinic exercises under the supervision of the rehabilitation veterinarian. Clinical signs indicating fatigue or relative overload of the cervical spine were discussed as signals to stop or change therapy, and daily rounds assured that the case was discussed with the team each time the patient visited for therapy. The animal chiropractor used manual therapies to maintain normal joint motion in the thoracolumbar spine and took time to counsel the owner about the condition. Communication with the primary care veterinarian was through biweekly updates. Communications also occurred after any significant re-evaluation and/or status change. Sophie did have intermittent episodes of pain on arising from a prone (down) position. These episodes were associated with higher levels of activity the previous day. Pain was managed with a non-steroidal anti-inflammatory and gabapentin. The owner was counseled about consistency of activity level. However, this was difficult in a



Figure 21.8 Custom cervical brace (Therapaw).

multi-dog household. Sophie improved in coordination slightly, and her strength improved markedly. She could turn more easily and had improved control of transitions. Sophie was transferred to a home exercise plan and coped well with no decline in clinical signs (Box 21.6, red flag box 4). She continued to see her chiropractor monthly and had rechecks at the rehabilitation clinic every 4–6 months.

Conclusions

Through a collaborative team approach, incorporating the strengths of each of the team members, best practices and best patient out-

comes are achieved in the rehabilitation setting. Regular communication, re-evaluation including measurement of objective outcomes, and respect for the rehabilitation chain of command and legal scope of practice of each team member is essential, especially when an unexpected problem or set-back is encountered. The overlap of the skill set provided by individual team members can be a bonus by allowing comprehensive patient care even when one team member is unavailable; or it can be a limitation resulting in competition, thus risking the integrity of the team. Focus on provision of the best possible rehabilitation practice/services while maintaining open communication will preclude these limitations.

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Equine Rehabilitation

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Introduction

The process of rehabilitating a horse after an injury or surgery is neither cookbook nor inflexible. Horses must be treated as individuals. The goal of a rehabilitation program is to achieve as close to the level of pre-injury physiologic and psychological fitness as is possible in the shortest amount of time.

Physiotherapy is common to human medicine. In the human field, there are many studies that have been performed on the different modalities that have demonstrated efficacy for specific conditions. An example would be the use of cryotherapy to decrease postoperative pain and swelling. However, there are some modalities that have limited or no proven efficacy.

In the horse, there is even less information available in the form of evidence-based medicine. However, research into the tissue effects and efficacy of therapeutic modalities has been increasing. The quality of the research that has been performed recently is improving though most are dealing with effects and not efficacy for a condition. Just because there is an effect does not mean a particular therapy is efficacious.

There is minimal work that has been done in the horse to validate the efficacy of different therapeutic modalities. Many of the treatment protocols have been adapted from human medicine for which there is better evidence of efficacy. There are clinical technique reports that address specific conditions or modalities, but these reports utilize techniques

that have been adapted from human medicine or personal experience. There is no reason not to use modalities that have been proved to be efficacious in the human, but efficacy determination needs to be performed in the equine.

Therapeutic Plan Development

A “shotgun” approach to developing and implementing a treatment plan should be avoided. It is important to realize that this is a team approach. The team consists of the veterinarian, veterinary technician/nurse, therapist, owner, as well as other professionals (trainers, farriers, etc.). All members of the team should be involved in all aspects of the treatment to ensure the best possible outcome.

There are several things one must look at while developing a rehabilitation program for an individual horse. The first is to have an understanding of success with patients that your team treated for a similar condition. How have they progressed during the treatment period? The second is to establish some outcome goals. Where should this patient be during a certain time in the rehabilitation process? Third, the team should determine therapist capabilities and the resources/facilities available. If a specific modality is indicated and you do not have it, then the patient should be referred to a clinic that has, or alternatively a different plan should be developed. Fourth, there should be a full assessment of the patient. What is the nature of the injury? What type of therapy or surgery has been done? How amenable is the patient to therapy? What are the owners’ expectations? These questions should be answered prior to initiating therapy. The team should develop a plan specifically for the patient based on the answers to the above questions.

Any plan will depend on the stage of the injury. The initial stage after injury is from the time of injury or surgery until all acute inflammation has resolved (Prentice, 2014).

This may be as short as a few days to as long as or longer than 3–4 weeks. The goal during this initial stage is to decrease pain and inflammation, preserve range of motion, and prevent muscle atrophy. Typically, cold therapy, supportive wraps, and passive motion are all utilized in this period.

The second stage after injury begins as the acute inflammation is resolving (Prentice, 2014). The goal of this stage is to gradually increase the stress being placed on the healing tissues. This aids in structural healing and preventing or revising scar tissue. Therapies that could be considered include therapeutic exercise, aquatic therapy, and modalities such as therapeutic ultrasound, shockwave therapy, as well as others.

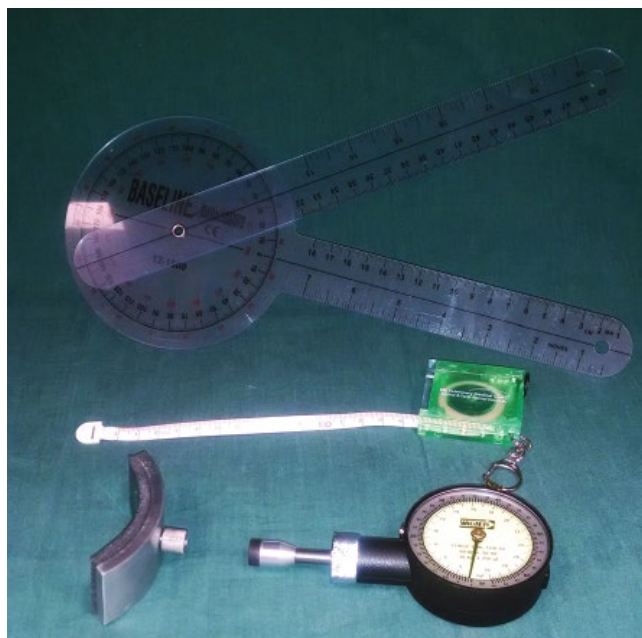
The choice of which modality to use in which stage is dependent upon the veterinarian, the therapists’ experience, and the exact goal for the modality. For instance, laser therapy may be used in stage 1 to help relieve tissue edema or inflammation and in stage 2 to help in epithelial migration during wound healing.

Therapeutic Monitoring

It is extremely important to develop methods for evaluating the effects of therapy and the response of the disease or injury to therapy – the distinction should be made between these two things. In addition, one must be able to document these changes. Ideally one should use quantifiable methods (Figure 22.1). There are fairly simple methods available. A tape measure may be used to measure limb circumference. Goniometers can be used to measure joint angles. Pressure algometry may be used to evaluate for lessening pain in an area. More sophisticated methods can also be used, including but not limited to radiographs, diagnostic ultrasound, thermography, or lameness evaluation (Paulekas and Haussler, 2009).

It is important that measurements be taken at the onset of therapy and frequent re-evaluations be performed. Early in the therapy period re-evaluations may need to be

Figure 22.1 Therapeutic monitoring equipment. From top to bottom: goniometer, tape measure, and pressure algometer.



performed daily or weekly, while later they may need to be done at monthly or longer intervals. If worsening or no improvement of the condition is noted the therapeutic plan must be modified.

Facilities, Equipment, and Personnel

Prior to the development of a rehabilitation practice it is important to read the State Veterinary Practice Act. In most states performing animal rehabilitation is considered the practice of veterinary medicine. As such, if a person is not a veterinarian then they will need to be a full-time employee of a veterinarian. If a person is a full-time employee of a veterinarian, then the level of supervision required will need to be determined. In most states a licensed veterinary technician/nurse can work under indirect supervision, meaning that the licensed veterinarian need not be on the premises, has given either written or oral instructions for treatment of the patient, is readily available by telephone or other forms of immediate communication, and has

assumed responsibility for the veterinary care given to the patient by a person working under his or her direction. However, laws vary from state to state and need to be reviewed for each state (province or country) in which a person is working.

It is important that the individuals developing rehabilitation plans, performing therapies, and monitoring progression should be thoroughly trained. This should be more in depth than the technical service training provided by equipment manufacturers. Individuals should seek out advanced training in the form of certification courses or by spending time training at a veterinary equine rehabilitation center. One of the major problems in the field of equine rehabilitation today is lack of formal training and education. Often therapies are provided that are of no benefit to the patient or in some cases are detrimental. Depending on what services are going to be provided, advanced training is available for equine rehabilitation, hyperbaric oxygen therapy, therapeutic taping, and massage, to name a few.

When planning a rehabilitation service, the demographics of the area needs to be evaluated.

Types of horses and their concentration in the area need to be determined. In addition, the equine disciplines (dressage, cross country, hunter/jumpers, reining, barrel racing, etc.), the types of clients, and the location of other rehabilitation facilities are a few other items that should be determined.

One also needs to decide whether the rehabilitation service is going to be ambulatory or haul-in to a clinic. If it is an ambulatory service, then this will limit the type of equipment and service you can provide. An ambulatory service will be limited to those modalities that can be easily transported in a vehicle and to prescribed therapeutic exercises, but is more economically feasible than a haul-in service. A haul-in service will be able to provide a much wider range of modalities but will also be a more significant expense.

Equipment needs will depend on all the factors presented above. For an ambulatory service, equipment that should be considered includes therapeutic ultrasound, therapeutic laser, thermal therapies, shockwave therapy, and equipment to perform therapeutic monitoring. For a haul-in facility, equipment in addition to that mentioned above includes underwater treadmill, swimming pool, a cold salt therapy unit, whole-body vibration, and pulsed electromagnetic therapy to name a few.

The above equipment need not be all obtained initially, but future needs should be anticipated. If funds are not available to purchase an underwater treadmill, for example, one should still plan for one in the future. Location and housing of the equipment need to be planned for. Most equipment is expensive, ranging from several hundred dollars to several thousand dollars. In addition to the purchase price of the equipment one must account for maintenance cost. In determining the charge for a service, you must take this into account in addition to the time it takes to perform the therapy. A fee must be charged that makes the service profitable. If you have a low number of cases, then you may have to charge a significant amount in

order to recoup the expenses. That is why it is extremely important to determine if the service area can support the equipment prior to purchase.

Nutrition

Nutrition is very important in the rehabilitation of the horse. Most horses that are painful and in the process of healing have specific nutritional requirements (Fleeman and Owens, 2007; Secombe and Lester, 2012). For example, horses are prone to the development of gastric ulcers. Practices that may be needed to reduce the incidence of ulceration is the feeding of a non-grain-based diet and alfalfa hay. An overly obese horse may only need grass hay fed at 2% of their body weight and a ration balancer. Horses that have laminitis or are prone to laminitis should be fed a low-starch feed and may need to have their hay soaked in water for 30–60 minutes to leach out the fructose (Longland *et al.*, 2011). Horses that are in a catabolic state will require additional calories. This can often be met by increasing the fat content of the diet instead of increasing the concentration portion of the diet. Lastly, all horse should have access to clean, fresh water and a white salt block or red mineral block.

Patient Environment and Mental Status

Many times, we focus on the injury and neglect the rest of the patient. We often have success in treating the injury, but lose use of the horse to contralateral limb problems such as laminitis, breakdown of supporting structures, and angular limb problems (in the case of foals and weanlings). Providing a high plane of nutrition, a good environment, and support to the other limbs is as important as the treatment of the injury. One should also not neglect the mental status of the horse. Having a companion in the barn (pony, another horse or goat), as well as play toys

(environmental enrichment) will go a long way in keeping the patient happy. Mental stimulation is important as many of the problems we are treating require long confinement of some type. Bedding type may need to be changed from straw to shavings or sand in the case of laminitic problems.

Therapeutic Modalities

Physiotherapy interventions that are commonly used in the horse include, but are not limited to, manual therapy, thermal agents, electrotherapeutic techniques, mechanical agents, therapeutic exercise, aquatic therapy, and hyperbaric oxygen therapy.

Manual Therapies

Manual therapies include massage, stretches, joint and soft tissue mobilization, and/or chiropractic (manipulation). These are used to restore optimum joint movement by reducing adhesions, mobilizing tight joint structures, and providing enhanced joint lubrication and joint nutrition.

Stretching and Massage

Stretching and massage can promote circulation, decrease muscle spasm, mobilize adhesions and scar tissue, and aid lymphatic drainage. They can also provide pain relief from tight muscles and connective tissue that are responsible for pressure or tension on nerve pathways, restore normal muscle length after injury and maintain normal muscle length (prevent shortening), avoid stiffness related to age or inactivity, and protect from stresses and strains.

Static or passive stretching consists of stretching a muscle (or group of muscles) to its farthest point and then maintaining or holding that position (Frick, 2010). This type of stretching is the most common type used in horses as the desired motion and positioning are controlled. It requires a human assistant to perform and is not under control of the horse. It is unclear how effective stretching is

in horses. No improvement in stride length was found after 8 weeks of stretching in a group of normal horses (Rose *et al.*, 2009), but they found several significant differences in joint range of motion between treatments in the shoulder, stifle, and hock. They concluded that the frequency with which passive stretches are applied to the horse appears to have some influence on horse movement. Their research did not demonstrate consistent improvement in equine movement because of passive stretching and highlighted the possibility that stretching on a daily basis may not be appropriate and may cause delayed onset muscle soreness. They suggested that the application of stretching on a three-times-per-week basis may be safer.

The basic science rationale for massage is supported by human research, indicating that massage may affect a number of physiologic systems as well as cellular and fascial components of the muscular system (Scott and Swenson, 2009; Hill and Crook, 2010). Equine therapeutic massage employs a number of techniques first developed in humans and has been reported to increase range of motion and stride length, reduce activity of nociceptive pain receptors, and reduce physiologic stress responses (Scott and Swenson, 2009). There is minimal evidence in the horses that suggest that massage is of any benefit other than for promoting lymphatic drainage (Buchner and Schildboeck, 2006). One study found that massage to the caudal limb muscles significantly increased passive and active limb protraction (Hill and Crook, 2010).

Myofascial Release

Myofascial release (MFR) is the use of the hands and fingers to apply pressure to cause a release of tension in muscle or fascia. It is a form of manual therapy that involves the application of a low load, long duration stretch to the myofascial complex, intended to restore optimal length, decrease pain, and improve function (Ajimsha *et al.*, 2015). MFR generally involves slow, sustained pressure applied to restricted fascial layers either

directly or indirectly (Ajimsha *et al.*, 2015). Direct MFR technique is thought to work directly over the restricted fascia: practitioners use knuckles or elbow or other tools to slowly sink into the fascia, and the pressure applied is a few kilograms of force to contact the restricted fascia, apply tension, or stretch the fascia (Ajimsha *et al.*, 2015). Indirect MFR involves a gentle stretch guided along the path of least resistance until free movement is achieved. The pressure applied is a few grams of force, and the hands tend to follow the direction of fascial restrictions, hold the stretch, and allow the fascia to loosen itself (Ajimsha *et al.*, 2015). The rationale for these techniques can be traced to various studies that investigated plastic, viscoelastic, and piezoelectric properties of connective tissue (Ajimsha *et al.*, 2015).

Mobilizations and Manipulations

These are aimed at pain relief, restoration of normal joint biomechanics and nerve function, improved muscle function, and promotion of healing. They are used to treat joint dysfunctions that limit range of motion by specifically addressing altered joint mechanics (Haussler, 2016). Factors that may alter joint mechanics include pain and muscle guarding, joint hypomobility (fixation), joint effusion, contractures, fibrosis or adhesions in joint capsules or supporting ligaments and degenerative joint disease.

Passive mobilization is a passive joint movement for increasing range of motion or decreasing pain (Goff, 2009). It is applied to joints and related soft tissues at varying speeds, amplitudes, or rhythm. The force is light enough that patient can stop the movement (Goff, 2009). Dynamic or active mobilization is a dynamic or active joint movement that is carried out and controlled by the patient (Goff, 2009).

Joint mobilizations and chiropractic manipulation are only to be performed by the rehabilitation veterinarian or animal physical therapist. Veterinary technicians/nurses can perform massage, passive range of motion exercises, and stretching techniques.

Manipulation is characterized by manual thrust delivered at high velocity in a specific direction. It incorporates a sudden, forceful thrust that is beyond the patient's control (Haussler, 2000). It is performed at the physiologic joint motion limit (elastic barrier) and into parapsysiologic space (Leach, 1994). It should only be done after receiving appropriate training.

Mobilizations have been evaluated in the horse. Most the studies have been directed toward the axial skeleton. Clayton *et al.* (2010a) found that the amount of bending in different parts of the cervical vertebral column differed among the dynamic mobilization exercises. As the horse's chin moved further caudally, bending in the caudal cervical and thoracolumbar regions increased, suggesting that the more caudal positions may be particularly effective for activating and strengthening the core musculature that is used to bend and stabilize the horse's back (Clayton *et al.*, 2010a). These researchers also determined that dynamic mobilization exercises performed in cervical flexion have applications in mobilizing the cervical and thoracic intervertebral joints (Clayton *et al.*, 2010a). Haussler *et al.* (2007) found that the passive vertical mobility of the trunk varied from cranial to caudal. At most sites, spinal manipulative therapy increased the amplitudes of dorsoventral displacement and applied force, indicative of increased vertebral flexibility and increased tolerance to pressure in the thoracolumbar portion of the vertebral column (Haussler *et al.*, 2007). In another study spinal manipulative therapy increased dorsoventral displacement of the trunk, which is indicative of producing increased passive spinal flexibility in actively ridden horses (Haussler *et al.*, 2010). Stubbs *et al.* (2011) found that dynamic mobilization exercise can cause enlargement of the multifidus muscle in as little as 3 weeks.

Based on the human literature, it has been recommended that mobilization for most injuries and post surgery can begin within 3 days if careful protocols are followed

(Schills and Turner, 2010). For severe ligament and tendon injuries, mobilization can begin at 3 weeks (Schills and Turner, 2010).

Thermal Agents

These include hot and cold applications. Heat or cold may be administered to horses using many modalities and can range from simply applying water from a hose to deep-heating ultrasound technologies.

Cold Therapy

The major physiologic benefits of cold therapy are decreases in circulation, cell metabolism, secondary tissue damage, edema, muscle spasm and pain (Belanger, 2010a). The benefits are most effective early in the period following injury or surgery. The primary effect of cold application is to constrict blood vessels. The reduced blood flow to tissues reduces edema, hemorrhage, and extravasation of inflammatory cells. Reduced tissue metabolism inhibits the effect of inflammatory mediators and slows enzyme systems. Analgesia follows cold therapy. The viscoelasticity of soft tissues is reduced with cold therapy.

Cold therapy is indicated in acute musculoskeletal injuries or inflammation and following surgical procedures to reduce edema, slow inflammation, and reduce pain. Cold is particularly effective during the first 24–48 hours after injury.

Cold may be applied by ice water immersion, application of ice packs, cold packs, salt water hydrotherapy units (Figure 22.2) and ice water-charged circulating boots.

Several studies have documented tissue effects or efficacy of cryotherapy in horses. The influence of hypertonic cold-water (5–9°C) spa bath hydrotherapy on the response of 27 horses with various lower leg injuries has been reported (Hunt, 2001). Fifteen horses with grade 2 or 3 superficial digital flexor tendon (SDFT) damage and four with suspensory ligament injury treated for 10 minutes three times a week responded with markedly improved ultra-



Figure 22.2 Cold salt water spa.

sonographic echogenicity and fiber realignment of injured tissues. All but two of these horses, when placed back into training returned to compete successfully within 6 months without re-injury. Petrov *et al.* (2003) were able to reduce the core temperature of the SDFT to a minimum of 10°C after 1 hour using a commercial compression splint with circulating coolant. These results indicated that topical application of cryotherapy can significantly reduce core SDFT temperature in standing horses and that the temperatures achieved *in vivo* during cold treatment were not detrimental to the *in vitro* viability of tendon cells. Cold immersion of the distal limbs is effective in reducing the severity of laminitis by decreasing the activity of laminar matrix metalloproteinases (MMP) and causing laminar vasoconstriction (van Eps and Pollitt, 2009; van Eps *et al.*, 2014).

Heat Therapy

The major benefits of heat therapy are increased local circulation, muscle relaxation, and increased tissue extensibility (Kaneps, 2000). By increasing blood flow tissue metabolites are mobilized, tissue oxygen levels are increased, and the metabolic rate of cells and enzyme systems are increased. Metabolic rate increases 2–3 times for a tissue temperature increase of 10°C. Increased blood flow and vascular permeability promotes resorption of edema. Heat application also decreases pain via the mechanisms cited earlier for cold therapy. Soft tissues may be stretched more effectively after warming. Heat is applied after acute inflammation has subsided. It is useful for reducing muscle spasms and pain because of musculoskeletal injuries. Heat therapy can be used to increase joint and tendon mobility. Heat may benefit recovery of localized soft tissue injuries by accelerating the healing response.

Superficial heat is most commonly applied using hot packs and hydrotherapy. Deep heat may be applied using therapeutic ultrasound. The most profound physiologic effects of heat occur when tissue temperatures are raised to 40–45°C (Kaneps, 2000). However, tissue temperatures above 45°C may result in pain and irreversible tissue damage (Kaneps, 2000). Heating for 15–30 minutes is required to elevate deep tissue temperature to the therapeutic range. Kaneps (2000) found that tissue temperature changes due to warm-water hose therapy ranged from 3.7 to 10.8°C. In addition, temperature elevation was short-lasting and dependent on the distance to the surface, with only limited effect on deep tissues more than 1.5–2 cm under the skin (Kaneps, 2000).

Using therapeutic ultrasound, Montgomery *et al.* (2013) found that the SDFT and deep digital flexor tendon (DDFT) are heated to a therapeutic temperature using a frequency of 3.3 MHz and intensity of 1.0 W/cm². The epaxial muscles are not heated to a therapeutic temperature using a frequency of 3.3 MHz and an intensity of 1.5 W/cm² (Montgomery *et al.*, 2013).

Electrical Techniques

These techniques include electrical muscle and nerve stimulation, pulsed electromagnetic field therapy, and low-level laser therapy (Schlachter and Lewis, 2016),

Neuromuscular Electrical Stimulation

Neuromuscular electrical stimulation (NMES) works by making the muscle contract through motor nerve stimulation using an interrupted direct current. Of all the electrotherapy techniques this is the most appropriate as it produces a controlled motor and sensory responses both superficial and deep, while maintaining a high level of compliance by the horse (Schills, 2009). Electrical muscle stimulation improves venous and lymphatic drainage, prevents muscle atrophy, prevents the formation of unwanted adhesions, reduces scar tissue formation, builds and re-educates damaged or weakened muscle, and encourages nutrition into the affected area.

Limited research is available on its use in the horse and it is conflicting. Bergh *et al.* (2010) used NMES on the m. gluteus medius and m. longissimus dorsi. They concluded that NMES treatment was well tolerated by the horse, but their protocol did not induce significant muscle adaptations. Ravara *et al.* (2015) concluded that functional electrical stimulation is a safe rehabilitation strategy in the management of equine epaxial muscle spasms and provides clinical improvements and some structural changes of the muscle tissue at the histological level (Box 22.1).

Transcutaneous Electrical Nerve Stimulation

Transcutaneous electrical nerve stimulation (TENS) is primarily used for pain modulation (Belanger, 2010b). It is the application of a pulsed electrical current over the skin surface for the purpose of pain modulation (Belanger, 2010b). There are no scientific studies in the current literature that document effect or efficacy in the horse. In the authors' experience it has not been found to decrease the need for analgesics in horses suffering from acute laminitis.

Box 22.1 University of Tennessee Equine Rehabilitation Department's protocol for neuromuscular electrical stimulation

Neuromuscular electrical stimulation is used primarily to treat or prevent muscle atrophy (Figure 22.3). The following is an example protocol for treatment of muscle atrophy:

- 1) Clip hair using a #40 blade where the electrodes are to be placed
- 2) Clean skin with alcohol
- 3) Apply electrodes to the area to be treated. Most commonly this will be at the origin and insertion of the affected muscle
- 4) Settings:
 - Waveform: Modified square wave with zero net direct current component
 - Contraction time: 10 s
 - Relaxation time: 30 s
 - Frequency: 30
 - Ramp: 3
- 5) Treatment protocol:
 - Once to twice daily
 - Start with 5 minutes, and then work up to 20 minutes (i.e., increase treatment

time by 2–3 minutes per day until 20–30 minutes is reached)

- It may take a few sessions to get a good contraction
- 6) Suggestions:
 - Start first session with 5 minutes, and then build up successive sessions, daily, by several minutes (2–3) to get to a 20- to 30-minute treatment session. It may help to allow the horse to eat hay during treatment for distraction
 - You should be able to see a good contraction while it is “on,” then a 10-second rest, then contraction again
 - Only adjust intensity up when “on”
 - You can assist with “active muscle” recruitment by imposing weight shifting onto the horse’s involved side when the unit is “on”
 - Economical, battery-operated units may be purchased and loaned to owners to provide treatment at home if needed

Pulsed Electromagnetic Field

Pulsed electromagnetic field (PEMF) therapy has been used to increase circulation, improve fracture healing, reduce pain and edema, and promote healing of wounds in humans and animals (Inoue *et al.*, 2002; Cinar *et al.*, 2008; Matic *et al.*, 2009; Strauch *et al.*, 2009; Rohde *et al.*, 2010; Nelson *et al.*, 2013). However, limited evidence is available in the horse that demonstrate effect or efficacy. It should be remembered that PEMF can have different intensities and frequencies depending on the unit. Biermann *et al.* (2014) evaluated the use of PEMF on back pain in polo ponies. They could not document an effect on either back pain or induced back movement. Rindler *et al.* (2014) also evaluated the effects of PEMF on the backs of actively working polo ponies. They found there was no measurable effect of PEMF on the

surface temperature of horses’ backs. Auer *et al.* (1983) reported beneficial effects for the repair of cortical stress fractures of metacarpal III (MC III) bones using PEMF, whereas Sanders-Shamis *et al.* (1989) did not find any significant differences in bone healing of surgically induced cortical lesions of MC III/metatarsal III between a PEMF and control group. Watkins *et al.* (1985) found that PEMF significantly delayed maturation of the tissue formed within the defect at weeks 8 and 12, as determined by histological examination. The collagen-type transformation was also delayed by PEMF, but not significantly. Collier *et al.* (1985) found that topical treatment with PEMF had no effect on the uptake of technetium in normal equine bone. The strongest evidence for efficacy with PEMF lies with non-union or slow-to-heal fractures (Schlachter and Lewis, 2016).



Figure 22.3 Neuromuscular electrical stimulation application for muscle strengthening.

Low-Level Laser Therapy

Low-level laser therapy (LLLT) (Figure 22.4), which uses an intense beam of light of a single wavelength, stimulates the body's processes, activates waste removal, increases repair activity, relieves swelling, heals surface wounds, and stimulates blood and lymphatic systems (Hamblin and Demidova, 2006). It also increases serotonin, thereby achieving a calming response (Hamblin and Demidova, 2006; Kazem Shakouri *et al.*, 2009; Silver, 2009). It has been reported that laser stimulation has certain biostimulating effects, such as accelerating cell division, increasing leukocytic phagocytosis; stimulating fibroblastic activity, and enhancing regeneration of lymph and blood vessels (Hamblin and Demidova, 2006; Kazem Shakouri *et al.*, 2009; Silver, 2009). Studies have shown that it also can cause vasodilatation



Figure 22.4 Low-level laser therapy application.

(Hamblin and Demidova, 2006; Kazem Shakouri *et al.*, 2009; Silver, 2009). These effects can assist wound healing and relieve chronic pain when properly applied (Hamblin and Demidova, 2006; Kazem Shakouri *et al.*, 2009; Silver, 2009).

Lasers are divided into safety classifications in the United States by the American National Standard Institute (ANSI). Classes 1, 2, and 3a have low power output and are not used for therapeutic purposes.

- *Class 1 lasers* or systems cannot emit accessible laser radiation more than the applicable class 1 accessible emission limit (AEL) for any exposure times within the maximum duration inherent in the design or intended use of the laser. Class 1 lasers are exempt from all beam-hazard control measures.
- *Class 2 lasers* are continuous wave (CW) and repetitively pulsed lasers with wavelengths between 0.4 and 0.7 μm that can

emit energy more than the class 1 AEL, but do not exceed the class 1 AEL for an emission duration less than 0.25 seconds and have an average radiant power of 1 mW or less.

- *Class 3a lasers* have an accessible output between 1 and 5 times the class 1 AEL for wavelengths shorter than 0.4 μm or longer than 0.7 μm , or less than 5 times the Class 2 AEL for wavelengths between 0.4 and 0.7 μm .
- *Class 3b lasers* cannot emit an average radiant power greater than 0.5 W for an exposure time equal to or greater than 0.25 seconds or 0.125 J for an exposure time less than 0.25 seconds for wavelengths between 0.18 μm and 0.4 μm , or between 1.4 μm and 1 mm. In addition, lasers between 0.4 μm and 1.4 μm exceeding the class 3a AEL cannot emit an average radiant power greater than 0.5 W for exposures equal to or greater than 0.25 seconds, or a radiant energy greater than 0.03 J per pulse.
- *Class 4 lasers* and laser systems exceed the class 3b AEL.

Research on the use of LLLT in the equine is limited. Bergh *et al.* (2007) found that minor superficial morphological changes occurred in equine epidermis after treatment with 91 J/cm² doses of defocused CO₂ laser and severe changes with a homogeneous eosinophilic acellular zone of underlying dermis and a significantly thinner epidermis, when irradiation doses of 450 J/cm² were applied. They also found that irradiation with defocused CO₂ laser causes a moderate to vigorous heating effect in superficial tissue, and a marked increase in blood flow (Bergh, 2006). The increase in temperature was of such intensity that there is a potential risk of thermal injuries to the skin (Bergh, 2006). The results also suggest that treatment with defocused CO₂ laser is not statistically better than placebo at reducing the grade of lameness in horse with traumatic arthritis of the fetlock joint (Bergh, 2006). Ryan and Smith (2007) found that light transmission was not affected by individual horse, coat color, or leg. However, it was associated with leg condition. They found that

tendons clipped dry and clipped and cleaned with alcohol were both associated with greater transmission of light than the unprepared state. Also, the use of alcohol without clipping was not associated with an increase in light transmission. Their results suggested that when applying laser to a subcutaneous structure in the horse the area should be clipped and cleaned beforehand.

The most prevalent method of indicating laser therapy dosage is to measure the density of energy applied to the tissue surface. This is typically expressed in joules per square centimeter (J/cm²). Some variation in clinical effects can be observed; particularly at very high (>50 W) or very low (<1 W) power settings using the same J/cm² dose. The dose required for a particular condition is dependent on the equipment used and what its wavelength is. In general, the more superficial a tissue the less energy required. In other words, the density of energy required for a wound will be less than that required for a tendon.

At the University of Tennessee's equine rehabilitation practice, the most common use of LLLT is to decrease pain and inflammation associated with musculoskeletal injuries and for wound healing.

Mechanical Agents

These agents include therapeutic ultrasound, shockwave therapy, and devices such as whole-body vibration units (Schlachter and Lewis, 2016).

Therapeutic Ultrasound

Therapeutic ultrasound is a form of acoustic energy used to treat musculoskeletal injuries, including inflammation and wounds. It offers deep heating without excessive heating of the skin. Ultrasound can also be used to decrease pain and muscle spasm, promote wound healing, aid reabsorption of hematoma, reduce swelling, and reduce scar tissue. It increases blood flow in the area treated. It increases cell membrane permeability to ions and other substances. It blocks signal transmission in

nerves. It decreases muscle spasms. It has been shown in clinical and scientific trials to increase collagen extensibility, enhance collagen remodeling, enhance collagen production, increase heat in deep tissues, increase blood flow, increase range of motion, reduce pain and muscle spasm, and accelerate wound healing.

The benefits of therapeutic ultrasound have been established in human and veterinary medicine (Dyson, 1990; Heckman *et al.*, 1994; Singh *et al.*, 1997; Doan *et al.*, 1999; Saini *et al.*, 2002; Fernandes *et al.*, 2003; Maiti *et al.*, 2006; Noble *et al.*, 2007). Therapeutic effects of ultrasound are both thermal and non-thermal (biologic). Thermal (tissue heating) effects are achieved by heating tissue using non-interrupted sound waves, or continuous ultrasound (Porter, 2005). Thermal effects result from energy carried by ultrasonic waves being attenuated and absorbed by tissue as the waves pass through it (ter Haar, 1999). Some of the positive effects of heat produced by therapeutic ultrasound include improved extensibility of collagen, decreased pain, decreased muscle spasms, and increased blood flow (ter Haar, 1999; Levine *et al.*, 2001; Noble *et al.*, 2007). Non-thermal (biologic) effects result from mechanical alteration of the local, cellular environment induced by the ultrasound waves (ter Haar, 1999; Siska *et al.*, 2008). To avoid heating the treated tissue and achieve non-thermal effects, pulsed ultrasound is used where pulse rates interrupt the sound waves at rates of 50%, 80%, or 90% (Porter, 2005). Changes in cellular environment may lead to modifications in cellular function resulting in a shorter inflammatory phase of healing, increased vascularity at the treatment site, and enhanced proliferation of fibroblasts (ter Haar, 1999; Levine *et al.*, 2001; Pounder and Harrison, 2008; Siska *et al.*, 2008). Non-thermal ultrasound has been used as an adjunctive therapy for patients with a fracture or a tendinopathy (Saini *et al.*, 2002; Fu *et al.*, 2008; Siska *et al.*, 2008).

The increase in temperature needed to achieve the desired therapeutic effect for thermal ultrasound has been established in people

(Draper *et al.*, 1995; Levine *et al.*, 2001; Demmink *et al.*, 2003). An increase in tissue temperature of 1°C is required to increase the metabolic rate of tissue, an increase of 2–4°C is required to lessen pain, muscle spasms, and chronic inflammation, and improve blood flow, and an increase >3°C is required to decrease the viscoelastic properties of collagen (Draper *et al.*, 1995; Levine *et al.*, 2001; Demmink *et al.*, 2003). For people, 1 MHz ultrasound is most effective at increasing temperature at a tissue depth of 2.5–5 cm, and 3.3 MHz ultrasound is most effective at increasing temperature at a tissue depth of 1.0–2.5 cm (Draper *et al.*, 1995). Results of a study performed to evaluate the effects of 3.3 MHz ultrasound on the temperature of the thigh musculature of dogs show that a 10-minute treatment at an intensity of 1.0 W/cm² increases the temperature of tissue by 2–4°C at depths of 1.0 cm and 2.0 cm. At an intensity of 1.5 W/cm², the temperature of tissue increases by at least 2–4°C at depths of 1.0 cm, 2.0 cm, and 3.0 cm (Levine *et al.*, 2001).

Research into the use of therapeutic ultrasound is lacking. Montgomery *et al.* (2013) found that the SDFT and DDFT are heated to a therapeutic temperature using a frequency of 3.3 MHz and intensity of 1.0 W/cm². However, the epaxial muscles are not heated to a therapeutic temperature using a frequency of 3.3 MHz and an intensity of 1.5 W/cm². Reis (2009), using a frequency of 1 MHz on pulsed mode, an intensity of 0.5 W/cm² for 5 minutes, treated collagenase-induced superficial flexor tendon lesions for 60 days. They found no significant difference clinically between treated and untreated limbs at either 15 or 60 days. However, there was a significant improvement in ultrasonographic evaluation at 60 days in treated limbs versus untreated. Singh *et al.* (1997) evaluated the use of ultrasound for 10 minutes per day at 1 W/cm² for 6 days after induction of arthritis in donkeys. They found that gross changes in the joint capsule, synovial membrane, and articular cartilage were mild in ultrasound-treated donkeys as compared to untreated controls.

At the University of Tennessee Equine Rehabilitation facility, therapeutic ultrasound is most commonly used for its heating effect on tendons and ligaments prior to exercise or mobilizations. The protocol used is 3.3 MHz and intensity of 1.0 W/cm^2

Extracorporeal Shockwave Therapy

Extracorporeal shockwave therapy (EWST) is a non-invasive treatment that involves the delivery of high-energy sound waves, or acoustic energy, to an affected area. Shock waves are characterized by high pressure bars of up to 100 bar and negative pressures of 5–10 bar (McClure *et al.*, 2003). They have a rapid rise time of 30–120 ns and a short ($5 \mu\text{s}$) pulse duration (McClure *et al.*, 2003). The mechanism of action seen in tissue and its relation to the clinical effect of treatment are not fully understood. Current theories include cytokine induction, increased osteoblast activity, stimulation of nociceptors, which in turn inhibits afferent pain signals, stimulation of neovascularization, induction of nitric oxide synthase and thus bone healing/remodeling, and induction of heat shock proteins (McClure *et al.*, 2000).

There are four methods for generation of sound waves: (i) piezoelectric; (ii) electromagnetic; (iii) electrohydraulic; and (iv) pneumatic-ballistic. Each has its own positive and negative traits. The first three are commonly referred to as focused extracorporeal shockwave units while the fourth is referred to as a radial extracorporeal shockwave unit. This is because of how the shockwave itself is generated.

EWST is generally considered safe (McClure *et al.*, 2003). No untoward effects have been noted on skin, tendons, ligaments, nerves, or bone (Boening *et al.*, 2000; McClure *et al.*, 2000, 2003). Mild skin irritation that resolved after 24 hours has been noted with radial ESWT (Boening *et al.*, 2000). Local analgesia lasting up to 3 days has been reported, resulting in the recommendation that horses should not undergo strenuous activity for a minimum of 4 days after treatment (McClure *et al.*, 2003). EWST is

not recommended for acute injuries. Sufficient time must be allowed for inflammation to subside and, in the case of tendons and ligaments, the lesion to organize.

Indications claimed to be effectively treated by EWST are insertion desmopathies, navicular syndrome, dorsal spinous process impingement, arthropathies (fetlock, pastern, and hock), exostosis, constriction of the annular ligament, tendinopathies with or without calcifications, calcification of the nuchal ligament, splint bone fractures, and dorsal metacarpal disease (Boening *et al.*, 2000; Crowe *et al.*, 2002; Palmer, 2002; McClure and Weinberger, 2003; Revenaugh, 2005; Waguespack *et al.*, 2011).

Treatment protocols are going to vary depending on whether focused or radial EWST is used and the particular manufacturer of the unit (McClure and Weinberger, 2003). As a rule, 1500–2000 pulses delivered to the affected site at 10- to 14-day intervals for 1–4 treatments are used. Horses are rested during the treatment period and, depending on the condition being treated, may require an extended rest period of up to 6 months.

Whole-Body Vibration Therapy

In whole-body vibration therapy (Figure 22.5), vibrating plates are used to induce oscillating vibrations throughout the body. It is most commonly used in human medicine to improve bone density and to improve neuromuscular function and muscle strength (Rehn *et al.*, 2006; Rauch, 2009; Slatkowska *et al.*, 2010; Von Stengel *et al.*, 2011; Chanou *et al.*, 2012; Pozo-Cruz *et al.*, 2012). However the literature is conflicting, with some reviews reporting benefit and others reporting none (Bautmans *et al.*, 2005; Cardinale, 2005; Nordlund and Thorstensson, 2006; Rehn *et al.*, 2006; Rauch, 2009; Feland *et al.*, 2010; Slatkowska *et al.*, 2010, 2011; Verschueren *et al.*, 2011; Von Stengel *et al.*, 2011; Wysocki *et al.*, 2011; Chanou *et al.*, 2012; Pozo-Cruz *et al.*, 2012; Costantino *et al.*, 2014). What is evident is that in those studies that do report benefits treatments



Figure 22.5 Whole-body vibration.

must be performed 2–3 times daily, 5–7 days per week for several months. In addition, to see improvement in neuromuscular function or muscle strength exercises must be performed while on the plate.

There are no scientific studies of whole-body vibration therapy in the horse. Thus, proof of effect and efficacy is based on antidotal evidence. Claims include improved circulation, increased bone density and muscle mass, reduction in injuries, and reduced stress. At the University of Tennessee Equine Rehabilitation Facility, whole-body vibration is used for those horses that are confined to a stall and cannot be adequately exercised or

have a limb immobilized. It is known that immobilization for several weeks causes a decrease in bone density (van Harreveld, *et al.*, 2002a, 2002b). Each case is placed on the plate for 20 minutes twice daily, 5 days per week.

Hyperbaric Oxygen Therapy

Hyperbaric oxygen therapy is the delivery of oxygen to the tissues by increasing environmental oxygen levels and pressure. Very little work has been done to evaluate hyperbaric oxygen therapy in horses and the work that has been done is conflicting. Dhar *et al.* (2012) found that hyperbaric oxygen treatment resulted in a significant increase in CD90-positive cells. Horses that did not yield any cells before treatment did so only after three hyperbaric oxygen treatments (Dhar *et al.*, 2012). However, Holder *et al.* (2008) found that the use of hyperbaric oxygen therapy after full-thickness skin grafting of uncompromised fresh and granulating wounds of horses is not indicated. Baumwart *et al.* (2011) found that pretreatment with hyperbaric oxygen prior to the administration of endotoxin did not offer any protection against endotoxemia.

Most indications and treatment protocols have been adapted from humans and research animals. More hyperbaric oxygen therapy units are now being placed in clinical practices and more information should be forthcoming. However, current clinical impressions are promising for certain conditions. The University of Tennessee Equine Rehabilitation Department is currently using hyperbaric oxygen therapy for treatment of wounds, osteomyelitis, acute laminitis, and postoperative colics, and has observed a more rapid improvement in these conditions versus those treated without hyperbaric oxygen therapy. Box 22.2 provides a list of indications for hyperbaric oxygen therapy.

Hyperbaric oxygen therapy works by increasing the amount of oxygen delivered to the tissues. This is accomplished by placing the horse in an enclosed chamber and increasing both the percentage of oxygen and

Box 22.2 Conditions that may benefit from hyperbaric oxygen therapy

- Thermal burns
- Smoke inhalation
- Cerebral edema
- Exceptional blood loss anemia
- Ileus
- Hypoxic encephalopathy
- Peripheral neuropathy
- Crush injuries
- Cellulitis
- Fracture healing – non-unions
- Delayed wound healing
- Compromised skin flaps and grafts
- Meningitis
- Refractory mycoses
- Carbon monoxide poisoning
- Cyanide poisoning
- Traumatic brain injury
- Gas gangrene – clostridial myonecrosis
- Cranial nerve paralysis
- Soft tissue swelling
- Compartment syndrome
- Acute and chronic osteomyelitis
- Tendon and ligament injuries
- Deep vein thrombosis
- Intra-abdominal abscess
- Neonatal maladjustment syndrome
- Tooth root abscess
- Myositis
- Neonatal septicemia
- Chronic sinusitis
- Laminitis

pressure within the chamber. Normal atmospheric pressure is 1 atmosphere absolute (ATA), which is equal to 14.7 pounds per square inch (PSI). Normal room air is approximately 21% oxygen. Most hyperbaric oxygen therapy treatments are carried out under 2–3 ATAs in the presence of 70–90% oxygen (Tibbles and Edelsberg, 1996; Gill and Bell, 2004; Slovis, 2008).

The amount of oxygen delivered to the tissues is dependent on the amount of oxygen in the air that is breathed, lung function, the amount of hemoglobin in the blood, and blood pressure (Slovis, 2008). Increasing the amount of air that is breathed does not improve the amount of oxygen delivery by hemoglobin, and breathing 100% oxygen at normal atmospheric pressure only increases the amount of oxygen in the blood by a small amount (Slovis, 2008). However, the partial pressure of oxygen in the blood is dramatically increased when breathing oxygen at two or more atmospheres of pressure. Increasing the atmospheric pressure during a treatment increases the solubility of oxygen into the body, allowing for an increased amount of oxygen to enter plasma and tissues (Slovis, 2008).

Oxygen can be considered a drug, in that not enough and too much can be detrimental

to the patient. Clinical use of oxygen under pressure should not exceed 3 ATA. No benefits are derived from further increases in pressure and the risk of toxicity increases (Gill and Bell, 2004; Slovis, 2008).

Hyperbaric oxygen can increase healing of hypoxic wounds (Slovis, 2008). Fibroblasts cannot synthesize collagen without adequate oxygenation (Slovis, 2008). More rapid wound healing occurs because of accelerated angiogenesis and prevention of excessive healing (granulation tissue formation). It can improve resistance to infection and accentuates the action of some antimicrobial agents (Tibbles and Edelsberg, 1996; Leach *et al.*, 1998; Slovis, 2008).

Hyperbaric oxygen can inhibit some clostridial toxins and increases the ability of polymorphonuclear leukocytes to kill *Clostridium perfringens* organisms (Slovis, 2008). It also has a direct killing effect on anaerobic organisms (Tibbles and Edelsberg, 1996).

Hyperbaric oxygen can lessen carbon monoxide toxicity and may be beneficial in animals suffering from smoke inhalation. It can increase osteoclastic activity and may reduce hematocrit and platelet aggregation, improving red blood cell stability. It also

improves the killing ability of neutrophils (Gill and Bell, 2004). Hyperbaric oxygen can decrease edema formation as evidenced by less edema following thermal injuries and ischemia (Tibbles and Edelsberg, 1996; Leach *et al.*, 1998; Slovis, 2008).

Hyperbaric oxygen therapy has complex effects on immunity, oxygen transport, and hemodynamics. The positive therapeutic effects come from a reduction in hypoxia and edema, enabling normal host responses to infection and ischemia (Gill and Bell, 2004).

Under normal circumstances, oxygen delivery in the body is dependent on the proportion of oxygen in the air that we breathe, lung function, the amount of hemoglobin in our blood, and the body's blood pressure. Increasing the amount of air that you breathe cannot improve the amount of oxygen delivery by hemoglobin, and breathing 100% oxygen at normal atmospheric pressure will only increase the amount of oxygen dissolved in blood plasma by a small amount. The partial pressure of oxygen in the blood is dramatically increased when breathing oxygen at 2 or more atmospheres of pressure. Increasing the atmospheric pressure in the chamber during a treatment increases the solubility of oxygen into the body. This process allows oxygen to enter plasma and tissues more readily and promotes the formation of new cells, thus further enhancing oxygen availability. Ischemic wounds and areas of infection now have new cells to increase circulation and supply life-saving oxygen and antibiotics.

Overall, hyperbaric oxygen therapy is safe. The Equine Rehabilitation Department has performed hundreds of sessions with minimal issues. Box 22.3 provides a list of contraindications. A strict protocol that includes "go or no go" evaluations is mandated. A checklist is utilized prior to and after therapy that includes animal evaluation as well as full assessment of the chamber function. All treatments are performed by individuals who have had advanced off-site training in hyperbaric oxygen therapy. It is critical that protocols be established and personnel are

Box 22.3 Contraindications for hyperbaric oxygen therapy

Absolute contraindications

- Untreated pneumothorax
- Tension pneumothorax
- Concurrent treatment with cis-platinum, disulfiram, or doxorubicin

Relative contraindications

- Chronic emphysema with CO₂ retention
- Optic neuritis
- History of spontaneous pneumothorax, thoracic surgery
- Seizure disorders
- High fevers
- Viral infections of the respiratory tract

adequately trained. Depending on the specific disease or injury, hyperbaric oxygen therapy can be the factor that pushes the patient toward a positive outcome when standard therapeutic measures are ineffective (Geiser, 2016).

Therapeutic Exercise

Therapeutic exercise is often used during the rehabilitation program. The amount and intensity is dependent upon the condition being treated, the extent of the damage, the time of healing and facilities available. Types of exercise include hand walking, riding, ponying (leading a horse while riding another), mechanical walker, underwater treadmill, swimming pool, and turnout to paddock or pasture. Each exercise program is tailored to the individual and may need to be adapted several times during the rehabilitation program. The basic principle is to reduce the force and strain on injured tissue while the normal reparative process proceeds (Davidson, 2016). Ground obstacles (ground poles, cavalettis) may also be incorporated to increase coordination and agility. In addition, change in terrain may be included to target specific areas (i.e., inclines to strengthen rear

limbs). The goal of therapeutic exercise to provide a gradual return to function, improve strength and coordination, and provide mental stimulation.

Equine Aquatic Therapy

Equine aquatic therapy primarily encompasses swimming and use of an underwater treadmill. Whirlpools and recovery pools are also examples but will not be addressed in this chapter. Equine swimming pools have been available for 25–30 years, but prior to that often ponds, lakes, or the ocean have been used. Many of the pools are located at tracks or barns. A few veterinary hospitals and equine rehab centers also have these pools. The main drawback of pools is the expense of construction and the costs of maintenance.

A cold salt water (hypertonic) spa has been reported to have anti-inflammatory, osmotic, and analgesic effects (Bender *et al.*, 2005). In horses, tendonitis and desmitis monitored ultrasonographically demonstrated reduced peritendinous and preligamentous edema, decreased inflammatory infiltration, and improved collagen fiber alignment after the 4 weeks of hypertonic cold-water therapy (Hunt, 2001). The added mineral components in water provide an increased osmotic effect, which reduces soft tissue inflammation and swelling, decreases pain, and ultimately improves joint range of motion. These osmotic effects can play an important role in managing soft tissue changes associated with musculoskeletal injury in horses (King, 2016).

More recently, equine underwater treadmills (Figure 22.6) have been developed to overcome the expense of construction and maintenance of in-ground pools. These units also provide a more controlled environment than do pools, with a decreased possibility of injury.

Swimming and underwater treadmill therapy may provide several benefits. They primarily provide cardiovascular conditioning without the stresses on the musculoskel-

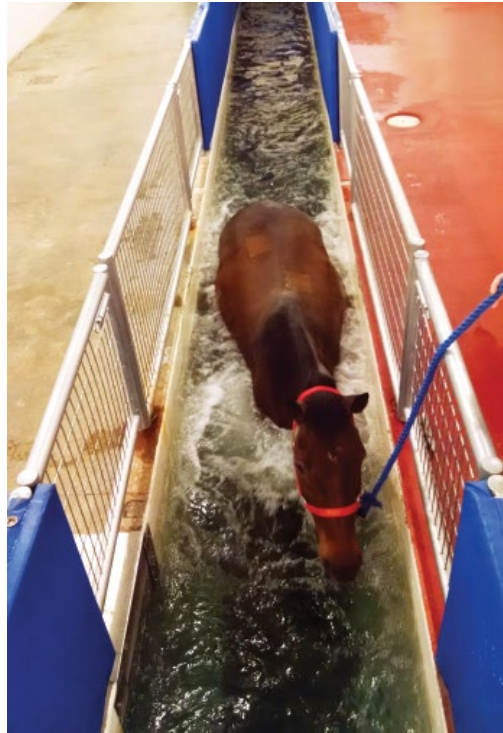


Figure 22.6 In-ground underwater treadmill.

etal system (Buchner and Schildboeck, 2006). In addition, they provide a different type of muscle exercise and also work different groups of muscles than when working on land. Resistance to joint movement is also a benefit that contributes to rehabilitation.

The use of swimming and underwater treadmill therapy in the rehabilitation of musculoskeletal injuries is becoming more common. Their use allows maintenance of cardiovascular fitness, muscle tone, and improved joint movement without undue stresses occurring on the injured limb. The reason for the decreased stresses placed on the limb is the buoyancy that the water provides, which depends on the depth of water and on the amount of water in relation to the body mass of the horse. Thus, if a horse is placed in a small area, less water will be required to get the water level to the point of the shoulder. Because of the reduced amount of water there is less buoyancy, thus more weight is borne by the horse.

Buoyancy is the force experienced as an upthrust, which acts in the opposite direction to the force of gravity (Monk, 2007). A body immersed in water appears to lose weight, and the weight loss is equal to the weight of water displaced (Monk, 2007). Immersion in water allows for unweighting of tendons, ligaments, bones, and joints within the distal limb (McClintock *et al.*, 1987). There is a reduction ground reaction forces, leading to reduced concussive stresses on joints and tendons, allowing for exercise without further trauma induced by weight bearing or concussive forces (Misumi *et al.*, 1994). Reduced body weight decreases post-operative and convalescent complications. McClintock *et al.* (1987) determined the weight reduction for a horse in a flotation tank filled with saline. They found approximately a 10% reduction in the weight borne by the limbs when the saline was at the level of the olecranon. When the saline was raised to the level of the tuber coxae there was approximately a 75% reduction in weight.

Immersion causes water displacement and increased hydrostatic pressure. Hydrostatic pressure is the sum pressure exerted on all surfaces of a body immersed in water, for any given depth (Monk, 2007). In humans this can cause redistribution of blood flow from the peripheral limbs due to an isotonic fluid shift from extravascular space (Yamazaki *et al.*, 2000). It can also lead to a decrease in hemoglobin and hematocrit level within 25–60 minutes of water immersion (Yamazaki *et al.*, 2000). Hydrostatic pressure will affect lung volumes, hence care needs to be taken with patients with respiratory distress or compromise (Monk, 2007).

Viscosity is the resistance of a fluid to motion (Monk, 2007). The viscosity of water increases as the speed of the horse increases. This is due to increased turbulence and drag, which in turn increases the amount and intensity of work being performed. The addition of hydrojets to the pool or treadmill can increase the drag on limb movement. Viscosity decreases as water temperature

increases (Monk, 2007). This means weaker and smaller muscles move more easily in warmer water (Monk, 2007).

The relative density and specific gravity of an object will depend on the composition of the object and determine whether an object will float or sink (Monk, 2007). Lean animals and heavily muscled animals, for example, have a tendency to sink and animals with a greater amount of body fat will float more easily (Monk, 2007).

Both swimming and underwater treadmill exercise are forms of aerobic exercise and help develop cardiovascular fitness. With water immersion there is a decrease in systemic vascular resistance and the changes in total peripheral resistance are dependent on water temperature (Yamazaki *et al.*, 2000).

Following underwater treadmill exercise there is a moderate but non-significant increase in blood lactate and plasma creatine phosphokinase levels (Lindner *et al.*, 2003). Hemoglobin concentration is significantly increased as a result of the physical exercise (Voss *et al.*, 2002). Voss *et al.* concluded that underwater treadmill training, following their training protocol, represents a medium-sized aerobic work load for horses.

Swimming causes a significant increase in blood pressure (Thomas *et al.*, 1980). However the maximum heart rate obtained while swimming is less than that obtained during ground exercise (Thomas *et al.*, 1980; Galloux *et al.*, 1994). There appears to be no relationship between heart rate and duration of swimming (Galloux *et al.*, 1994). There also appear to be increased cardiovascular benefits while working at slower speeds (Galloux *et al.*, 1994; Hobo *et al.*, 1998).

Water pressure on the horse's body during swimming prevents adequate ventilation (McClintock *et al.*, 1986a, 1986b; Hobo *et al.*, 1998). Hobo *et al.* (1998) found an increase in respiratory rate, an increase in both inspiratory and expiratory pressure, and that the expiratory time was roughly doubled the inspiratory time (Hobo *et al.*, 1998). This suggested that a longer expiratory time may limit sudden collapse of airways by water

pressure during swimming and prevent a marked decrease in air space volume, thus maintaining buoyancy (Hobo *et al.*, 1998). There are no studies available on the effects of underwater treadmill exercise on the respiratory function in horses.

Walking in water at the level of the carpus or ulna resulted in a lower stride frequency and greater stride length compared to walking in water at hoof height (Scott *et al.*, 2010). Water provides a resistance to movement of the limb in the sagittal plane, so an increase in height of the flight arc may also minimize the resistance experienced in swinging the limb back and forth (Scott *et al.*, 2010). When moving in water between carpal and ulna height, the horse may find it easier to adopt a rounder flight arc by increasing flexion of the hip, stifle, and hock joints. Water treadmill exercise may increase activity of muscles which flex the hip, flex the stifle, and protract the hindlimb (Scott *et al.*, 2010). Borgia *et al.* (2010) found no effect of water treadmill training on the properties of the gluteal and superficial digital flexor muscles and on cardiocirculatory response to a standardized exercise test. However the authors state that a more strenuous water treadmill conditioning protocol may be needed to induce a training effect in gluteal and superficial digital flexor muscle and heart rate response Borgia *et al.*, 2010).

Evaluating the effects of swimming on 2-year-old thoroughbreds in race training, Misumi *et al.* (1995) found that fast twitch, high oxidative fibers increased. There was an increase in aerobic capacity of muscles and a decrease in fast twitch, low oxidative fibers. There was no change in slow twitch fibers. They suggested that a training program including swimming training is seen as being useful for improvement in performance capacity, since it can reduce locomotor diseases in young horses and allow for smooth progress in future training (Misumi *et al.*, 1994, 1995).

Horses use their forelimbs to regulate their lateral balance and their rear limbs function in propulsion (Galloux *et al.*, 1994). The propulsive

action of the rear limbs is much exaggerated. During swimming, the equine spine is lordotic. Because of the exaggerated rear limb action and lordotic back, horses with rear limb injuries or back pain should not swim.

There is an increased range of joint motion in both fore- and hindlimbs depending on water height. Joint angle in horses decreases as water approaches the carpus or hock. Once the level of the carpus or hock is reached, joint flexion and limb height will vary little. This may be used to target specific joints and aid in re-establishment of joint range of motion after joint surgery. In dogs, there is increased flexion and range of motion during swimming compared to walking in both normal and operated stifle joints post-cranial cruciate ligament surgery (Marsolais *et al.*, 2003). The increased range of motion was due to increased joint flexion. Ground treadmill walking produces greater stifle extension than swimming (Marsolais *et al.*, 2003).

Water density is 12 times greater than that of air (Monk, 2007). During aquatic therapy there is increased resistance to limb or body movement and increased energy costs compared to walking at similar speeds on land due to this increased density. This provides better muscle development and muscle tone due to working against resistance and provides better balance of muscle groups working against increased resistance while maintaining a symmetrical gait (Bromiley, 2000). In addition, underwater treadmill exercise significantly improves the horse's postural stability (King *et al.*, 2013).

Safety is a paramount importance when using swimming or an underwater treadmill. It is very important that the handlers are thoroughly familiar with the equipment and able to read the horse's temperament. The handler should be able to anticipate and correct problems with the horse or equipment before they develop.

Swimming pools should be constructed so that two handlers can easily walk 360 degrees around. Depth should be adequate so that the horse cannot touch the bottom. Most are 12–15 feet (3.5–4.5 m) deep. The sides should

be sloped to prevent injury. Some type of ramp system should be employed that allows easy entry and exit from the pool. The filtration system is very important. The water becomes quickly contaminated with dirt and feces so a good filtration system is a necessity. Most horses are good swimmers but do require training. Usually an introductory period is required with increasing time intervals in the pool. The time is slowly increased to a period of approximately 15 minutes. Little research has been done on proper protocols. There has been a protocol described using a swimming test to determine the level of fitness a horse has achieved following conventional training. However, it is not very applicable for determining swimming protocols.

There are two types of underwater treadmills: in-ground and above-ground. The in-ground type allows for a larger volume of water to be used and so there is greater buoyancy. Both have variable speeds that range up to 15 mph (24 km/h). This is much lower than the high-speed treadmill, which can achieve speeds of 45–50 mph (72–80 km/h). Some may require sedation until they are familiar with the routine. Both units require filtration and most can provide both heated and unheated water. These units are expensive but the prices are becoming more reasonable.

Acclimation to water treadmill exercise requires a minimum of two 15-minute non-sedated acclimating runs (Nankervis and Williams, 2006). Sedation can be used to prevent horses panicking during the first exposure but, thereafter, does not affect the time taken to acclimate.

Water temperature should be adjusted to provide maximize comfort. For active exercise and swimming use 18–27 °C (65–82 °F). For less vigorous exercise 35–40 °C (96–104 °F) is acceptable. The least adverse physiologic effects occur at 36 °C (97 °F).

Unfortunately, most of the facilities do not monitor any parameters either during or after a session. Because of this, recommended protocols are empirical. Two easy parameters that can be measured include heart rate and blood lactate levels. Heart rate can be used to

monitor the level of stresses that are being placed on the cardiovascular system. Maximum heart rate in the horse is approximately 200 bpm. By using a heart rate monitor you can set a target rate and then set a time to stay in the target range. Blood lactate is used to determine if the horse has progressed into the aerobic metabolism stage. This is a desired state to achieve adequate conditioning. Respiratory rate can also be used but is not as reliable as heart rate. In addition, lameness evaluation should be performed at weekly intervals to determine if any musculoskeletal problem has arisen.

Indications for aquatic therapy include rehabilitation after injury or surgery, tendon injuries, post-arthroscopic surgery, replacement for hand walking, joint stiffness, osteoarthritis, increase in muscle development, encourage symmetric gait and back development, cardiovascular conditioning, and reconditioning after a lay-up.

Contraindications for aquatic therapy include acute joint inflammation, skin infections, open wounds, upper limb lameness (swimming), back pain (swimming), acute myositis, cardiovascular compromise, and respiratory disease.

The following are variables that must be considered when developing a protocol for an individual patient. Of utmost importance is the disposition of the horse. There are going to be some horses that never get used to aquatic therapy. In these individuals, another therapeutic plan will need to be developed.

- Injury and condition of patient
- Water level (if possible)
 - Amount of buoyancy and limb weight bearing
 - Degree of joint flexion desired
- Water temperature: warm versus cold
- Treadmill speed
- Hydrojets – on or off (if equipped)
- Warm-up period
- Duration of exercise: 5–30 minutes
- Exertion during exercise
- Frequency of exercise
- Cool-down period.

Box 22.4 Sample underwater treadmill program**Acclimation period (days 1 and 2)**

- Walk in and walk out of underwater treadmill
- Walk in unit and add 6 inches (15 cm) of water, drain, and walk out to acclimate to sounds of unit filling
- Walk in, add water, turn on treadmill, stop treadmill, drain unit, and walk out
- May use sedation during acclimation if needed. It also helps to have a trained horse at the facility to act as a lead horse

Rehabilitation program days 3–7

- Water level at olecranon
- Speed – walk at 2 mph (3 km/h)
- Warm-up – 5 min at 2 mph (3 km/h)
- Duration of active walk – 5 min at 2 mph (3 km/h)
- Cool-down – 5 min at 2 mph (3 km/h)
- Frequency – Once per day; 5 days per week
- Outcome measures:
 - Walking comfortably for 5 min at 2 mph (3 km/h)
 - If successful proceed to next level

Rehabilitation program week 2

- Warm-up – 5 min at 2 mph (3 km/h)
- Duration of active walk – 10 min at 3 mph (5 km/h)
- Cool-down – 5 min at 2 mph (3 km/h)
- May begin lowering or raising water level
- May increase speed to 2–3 mph (3–5 km/h)
- Frequency – Once per day; 5 days per week
- Outcome measures:
 - Walking comfortably for 10 min at 3 mph (5 km/h)
 - If successful proceed to next level

Rehabilitation program week 3

- Warm-up – 5 min at 2 mph (3 km/h)
- Duration of active walk – 15 min at 4 mph (6.5 km/h)
- Cool-down – 5 min at 2 mph (3 km/h)
- Adjust water level to best address specific injury and goals
- Frequency – Once per day; 5 days per week
- Outcome measures:
 - Walking comfortably for 15 min at 4 mph (6.5 km/h)
 - If successful proceed to next level

Rehabilitation program week 4

- Warm-up – 5 min at 2 mph (3 km/h)
- Duration of active walk – 20 min at 5 mph (8 km/h)
- Cool-down – 5 min at 2 mph (3 km/h)
- Frequency – Once per day; 5 days per week
- Outcome measures:
 - Walking comfortably for 20 min at 5 mph (8 km/h)
 - If successful proceed to next level

Rehabilitation program week 5

- Warm-up – 5 min at 2 mph (3 km/h)
- Maximum exercise intensity of 5 mph (8 km/h) for 20–30 min
- Cool-down – 5 min at 2 mph (3 km/h)
- May introduce cross-training activities
- Frequency – Once per day; 3–5 days per week;
- Outcome measures:
 - Walking comfortably for at least 20 min at 4–5 mph (6.5–8 km/h)

Sample Underwater Treadmill Protocol Box 22.4 gives an example of an underwater treadmill program for a mild to moderate tendon injury. It is only begun after an ultrasound evaluation has shown significant healing has occurred and that the horse can have a significant amount of hand walking. It is

important that the horse be evaluated prior to each treadmill session for any increased heat or swelling of the affected tendon or ligament. It also should be evaluated weekly for change in the degree of lameness. Ultrasonography should be repeated at monthly intervals so that the rehabilitation

program can be adjusted or terminated if necessary. Note that the ability to vary water height to any appreciable amount only applies to above-ground units. If an in-ground treadmill is used, then disregard the water height recommendation. All sessions should have a 2- to 5-minute warm-up and 2- to 5-minute cool-down periods.

Ground Exercises

Ground exercises are exercises that are performed at hand or while being ridden. They are usually targeted at a specific area, such as improvement in proprioception and coordination, strengthening of specific muscles, improving joint mobility, and improving overall body condition. There are many types of exercises that may be used. It is often based on the imagination of the rehabilitation veterinarian, rehabilitation physical therapist and rehabilitation veterinary technician/nurse to develop specific exercises. One must consider the nature of the patient and injury, the equipment and facilities available, and the personnel. Many of the therapeutic exercises are inexpensive and just require time and personnel.

Proprioception and coordination may be improved by walking over ground poles placed randomly (Figure 22.7). Other configurations and different obstacles can be used (Paulekas and Haussler, 2009). Different surface transitions may also be tried, such as going from grass, to sand, to gravel, to water, and then to asphalt. Joint mobility can be



Figure 22.7 Ground cavaletti poles laid in random fashion to improve proprioception.

worked on by having cavalettis arranged like spokes on a wagon wheel and having the horse step over them, going in both directions (Figure 22.8).

Bracelets or weights can be placed around the pasterns (Clayton *et al.*, 2010b, 2011a, 2011b). However, horses will become habituated to the devices (Clayton *et al.*, 2008). Core muscle stability can be improved utilizing therapeutic bands placed around the caudal limbs or abdomen (Paulekas and Haussler, 2009). These may be used at hand, lunging, or while being ridden (Figure 22.9).

Strengthening of rear limb musculature may be accomplished by riding up and down gradual inclines or pulling a cart. This most commonly has been used as a therapy for intermittent upward fixation of the patella.

Kinesiology Taping

Kinesiology taping is an elastic therapeutic tape method used for treating sports injuries and a variety of other disorders. A chiropractor, Dr. Kenso Kase, developed these taping techniques in the 1970s (Williams *et al.*, 2012). It is claimed that it supports injured muscles and joints and helps relieve pain by lifting the skin and allowing improved blood and lymph flow (Williams *et al.*, 2012) (Figure 22.10).

There is conflicting evidence in humans as to whether kinesiology taping is effective or not (Williams *et al.*, 2012; Montalvo *et al.*, 2014; Parreira *et al.*, 2014). There is no scientific



Figure 22.8 Ground cavaletti poles laid in a radius to encourage joint motion and improve coordination.

Figure 22.9 Therapeutic band application to improve core muscle strength.



Figure 22.10 Equine kinesiology taping.

evidence to support its use in the equine. The effect and efficacy of kinesiotaping is anecdotal or based on human literature. Courses

are available in the proper technique of equine kinesiology taping.

Return to Work

At some point the rehabilitation program will end and the horse must be returned to work if possible. It is important that return to work is done gradually. Although the particular injury may have healed, the rest of the body may not be in condition. Table 22.1 gives an example of a conservative back-to-work plan. We always recommend at least a 10-minute warm-up and cool-down period.

Conclusion

The equine rehabilitation veterinary technician or nurse plays an important role in carrying out the rehabilitation equine veterinarian's treatment plan and communicating that plan to the owner on a routine basis. Monitoring the patient's progress and maintaining the records will be important to track trends for the patient. Monitoring pain and advancement in achieving the therapeutic goals will be necessary to aid the rehabilitation veterinarian and physical therapist in developing plans. By understanding all aspects of equine physical rehabilitation, the equine physical rehabilitation veterinary technician or nurse will contribute to the success of bringing an injured patient back to the goals set forth by the team.

Table 22.1 Example back-to-work plan following resolution of injury.

Initially start with 3 days of riding. After 2nd week can be increased to 5 days per week if desired						
Week	Walk	Trot	Walk	Trot	Walk	
	20 min walking under saddle					
2	30 min walking under saddle					
3	40 min walking under saddle					
4	10 min	2 min	6 min	2 min	10 min	
5	10 min	3 min	4 min	3 min	10 min	
6	10 min	4 min	2 min	4 min	10 min	
7	10 min	6 min	2 min	6 min	10 min	
Starting week 8 add canter						
Week	Walk	Trot	Canter	Walk	Trot	Walk
8	10 min	2 min	2 min	5 min	2 min	10 min
9	10 min	3 min	2 min	4 min	3 min	10 min
10	10 min	2 min	3 min	4 min	2 min	10 min
11	10 min	4 min	4 min	4 min	4 min	10 min
Weeks 1–6						
Start turning out into a paddock/round pen. Start with 2 hours per day and build up to approximately 8 hours by week 4. Can also be hand walked 20–30 minutes, 2–3 times per day during this time if desired (in addition to walking under saddle).						
Weeks 7–11						
Turn out to pasture; start with 2 hours per day of pasture and 6 hours of paddock/round pen. Increase pasture time by 2 hours weekly so that by week 11 he or she is on pasture at least 8 hours per day.						
Starting week 12						
No restrictions in work or turn out; resume normal work over a 2- to 3-week period. Recommend re-evaluation prior to canter work and again prior to resuming regular work.						

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23

Adjunctive Therapies Part 1: Acupuncture and Traditional Chinese Medicine

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Traditional Chinese Veterinary Medicine Basics

Brief History

Traditional Chinese Veterinary Medicine (TCVM) is a medical system that has been used in China to treat animals for about 2000 years. TCVM emerged from its parent system, Traditional Chinese Medicine (TCM) that is used to treat people. Over the years, TCVM has become widely accepted in the

United States and in Europe. Many of the original theories are still used today; however, with advances in modern medicine new treatment techniques have been employed. TCM is categorized into five branches: acupuncture, Chinese herbal medicine, tui-na massage, food therapy, and qi gong. Due to the difference in veterinary patients, the only branch that is vastly different in TCVM is the fifth branch. This branch has been termed “exercise,” which can be considered as physical rehabilitation.



Figure 23.1 Dr. Carolina Medina places acupuncture needles in JimmyCat (Mary Ellen Goldberg's cat).

Five Branches of Traditional Chinese Veterinary Medicine

Acupuncture

Acupuncture is defined as the stimulation of a specific point (acupuncture point) on the body with a specific method, resulting in a physiologic effect. These physiologic effects include both systemic and local effects as acupuncture stimulates both the central and peripheral nervous systems. Some of the systemic effects include release of endogenous substances such as beta-endorphins (Han *et al.*, 1984; Wu *et al.*, 1995; Skarda *et al.*, 2002), dynorphins, enkephalins, serotonin (Costa *et al.*, 1982; Scherder and Bouma, 1993), epinephrine, gamma-aminobutyric acid (GABA), cortisol, and various hormones. Acupuncture can also improve blood flow to the pituitary axis and capillary wall enzyme concentration; release somatotropin in chronic pain; induce luteinizing hormone release which triggers ovulation; stimulate prolactin release to promote lactation; stimulate oxytocin release to induce uterine contraction; and modulate thyroid function. The local effects of acupuncture include muscle relaxation and spasm relief; release of Hageman's factor XII, which activates the clotting cascade, complement cascade,

plasminogen, and kinins; degranulation of mast cells, which releases histamine, heparin, and kinin protease; release of bradykinin, which leads to vasodilation; and production of local prostaglandins, which leads to smooth muscle relaxation (Smith, 1994) (Figure 23.1).

Chinese Herbal Medicine

Pioneer TCVM practitioners used individual (single) Chinese herbs to treat diseases. Over time they found that combining several Chinese herbs into a formula was more effective than single herbs alone. A Chinese herbal formula contains different quantities of several herbs. A typical Chinese herbal formula contains anywhere between 4 and 15 herbs, each of which has an assigned a role and mechanism of action within the formula. Formulas consist primarily of plant particles including bark, stems, roots, flowers, and/or seeds. There are a few formulas that contain minerals and animal-origin products.

Tui-na Massage

Tui-na is a form of Chinese manual therapy used for prevention and treatment of disease. It incorporates massage, acupressure, and physiotherapy techniques. It can be applied

directly to acupuncture points, trigger points, and meridians. In Chinese, Tui-na means “push-pull and lift,” which is the basis of most of the techniques. Tui-na can be very useful as it is practical to apply to many cases. It can be performed by the veterinarian and/or technician and certain simple techniques can be taught to owners to perform at home. Like massage, Tui-na increases local circulation and leads to vasodilatation, which warms the soft tissues and provides pain relief.

Food Therapy

Food therapy is the use of the healing power of food and is often utilized alone or more frequently used in conjunction with other TCVM therapies to treat disease. Western medicine has lately begun to understand the effect of food on healing the body; the interaction between genes and nutrients is termed “nutrigenomics” (Sales *et al.*, 2014). TCVM food therapy is the art and science of tailoring diet plans to individual patients and food ingredients are chosen based on TCVM principles. Like other TCVM modalities, the ultimate goal of food therapy is to restore and maintain balance in the body. However, given its very nature, the effects of food therapy are slower acting than modalities such as acupuncture and herbal medicine. On the other hand, there are virtually no side-effects when food ingredients are chosen correctly, and food therapy is a mode of treatment that can be used safely throughout a patient's lifetime.

Exercise/Qi Gong

Qi gong is the practice of aligning body, breath, and mind for health. It is traditionally viewed as a practice to cultivate and balance Qi (“life energy”). Qi gong typically involves meditation, coordinating slow flowing movement, deep rhythmic breathing, and a calm, meditative state of mind. Since veterinary patients are incapable of practicing qi gong, exercise in the form of physical rehabilitation is considered a suitable substitute.

Yin Yang Theory

The yin yang theory depicts everything as being composed of two opposing yet complementary pairs of opposites. The simplest example is the sun and the moon, in which the sun is considered yang while the moon is considered yin. Anything that relates to activity, brightness, or function is considered yang, while anything that relates to inactivity, darkness or structure is considered yin. In regards to medicine, physiological activities belong to yang while nutrient substances correspond to yin.

The yin yang theory is incorporated into TCVM to assist in the diagnosis and treatment of disease. According to TCVM, disease does not occur if yin and yang are in balance. There are four possible states of imbalance: excess yin, excess yang, yin deficiency, and yang deficiency. Excess yin typically has an acute onset of clinical signs with a short course of disease and can present with edema, pain, and/or loose feces. Excess yang also has an acute onset of clinical signs with a short course of disease but presents with a high fever and/or hyperactivity. Common signs of yin deficiency include general weakness, polydipsia, low grade fever, and the disease state is generally chronic in nature. Yang deficiency is seen in patients with a chronic history of urinary incontinence, low back pain, limb edema, and cold extremities (Xie, 2005).

Five Element Theory

The five elements refer to five categories of nature: wood, fire, earth, metal, and water. There is an enhancing, inhibiting, and restraining relationship between them to maintain a balance. Like the yin yang theory, the five element theory is embodied into TCVM to assist in the diagnosis and treatment of disease. The five element theory is used to describe the nature of the anatomical structures and their relationships with each other, in addition to the relationship

between the patient and their environment. The liver, gallbladder, eyes (vision), tendons, ligaments, and nails belong to the wood element. Therefore, a patient with a cranial cruciate ligament tear is considered to have a disharmony with the wood element. The cardiovascular system, including the heart and pericardium, and the small intestines and tongue (voice) correspond to the fire element. A patient with congestive heart failure would have a disharmony with the fire element. The spleen, stomach, muscles, and mouth (taste) are related to the earth element, causing a patient with gastrointestinal disease to suffer from an earth element imbalance. The lungs, large intestines, skin, hair coat, pores, and nose (smell) are associated with the metal element. An atopic dermatitis case would be classified as a metal element disharmony. The kidneys, bladder, bones, reproductive system, and ears (hearing) belong to the water element. A patient with osteoarthritis would be categorized as having a water element disharmony (Xie, 2005).

Acupuncture Points

Research shows that most acupuncture points are in areas on the skin of decreased electrical resistance or increased electrical conductivity (Urano and Ogasawara, 1978). This can be measured by using a point finder, acupoint detector, or AC dermometer. In addition, it has been found that acupuncture points are closely associated with free nerve endings, veins, lymphatics, and an aggregation of mast cells (Jaggar and Robinson, 2001). There are 173 major acupuncture points in horses and 361 points in people. Acupuncture points correspond to four known neural structures. Type I acupuncture points, which make up 67% of all points, are considered motor points. The motor point is the point in a muscle which, when electrical stimulation is applied, will produce a maximal contraction with minimal intensity of stimulation. Motor points are in areas where

nerves enter muscles. For instance, SI-9 is located at the junction of the deltoid muscle and triceps brachii and is supplied by axillary and radial nerves. Type II points are located on the superficial nerves in the sagittal plane on the dorsal and ventral midlines. For instance, *Bai hui* lies in the depression between the spinous processes of the seventh lumbar and the first sacral vertebrae on dorsal midline and is supplied by the dorsal branch of the last lumbar nerve. Type III points are located at high-density loci of superficial nerves and nerve plexuses. For example, GB-34 is located at the point where the common peroneal nerve divides into the deep and superficial branches cranial and distal to the head of the fibula. Type IV points are located at the muscle–tendon junctions where the Golgi tendon organs are located. For example, BL-57 is located at the junction between the gastrocnemius muscle and the calcaneal tendon (Gunn, 1997; Hwang and Egerbacher, 2001).

Meridians

Most commonly, acupuncture points are located on meridians. Meridians are channels or conduits of energy that connect the acupuncture points throughout the body. There are 14 major meridians, named Lung (LU) Meridian, Large Intestine (LI) Meridian, Stomach (ST) Meridian, Spleen (SP) Meridian, Heart (HT) Meridian, Small Intestine (SI) Meridian, Bladder (BL) Meridian, Kidney (KID) Meridian, Pericardium (PC) Meridian, Triple Heater (TH) Meridian, Gallbladder (GB) Meridian, Liver (LIV) Meridian, Governing Vessel (GV) Meridian, and Conception Vessel (CV) Meridian. A close correlation exists between meridians and peripheral nerve pathways. Meridians possess bioelectric functions like peripheral nerves and they follow along peripheral nerves. For example, the Lung Meridian follows the musculocutaneous nerve and the Pericardium Meridian follows the median nerve. In a study to investigate the locations of meridians,

researchers injected a radio-isotope into one acupuncture point and visualized it gradually accumulate at another acupuncture point on the same meridian. Radio signals over one acupuncture point can be picked up at another acupuncture point on the same meridian. Meridians conduct current, with flow toward the central nervous system (Smith, 1994).

Methods of Stimulation

Acupuncture can be stimulated by various means, including acupressure, dry needle, electro-acupuncture, aqua-acupuncture, moxibustion, gold implantation, pneumo-acupuncture, hemo-acupuncture, and laser acupuncture (Altman, 1994; Ferguson, 2007).

- In acupressure, firm digital pressure is applied to an acupuncture point for a specific length of time (e.g., 5 minutes). This is the least invasive type of stimulation.
- Dry needle is the insertion of a sterile filiform needle into an acupuncture point to elicit a response. These needles are typically left in place for 15–30 minutes. Dry needle is the most commonly used technique in veterinary medicine (Figure 23.2).
- In electro-acupuncture, electrical currents are applied to the dry needles to increase the therapeutic response (Figure 23.3).

This modality is desirable because you can adjust the frequency and amplitude and induce a stronger stimulation than dry needle alone. Low-frequency (1–40 Hz) electro-acupuncture predominantly stimulates A-delta fibers and releases beta-endorphins and met-enkephalins (Figure 23.4). High-frequency (80–120 Hz) electro-acupuncture predominantly stimulates C fibers and releases dynorphins. Very high-frequency (200 Hz) electro-acupuncture predominantly stimulates serotonergic fibers and releases serotonin and epinephrine (Smith, 1994).

- Aqua-acupuncture is the injection of a sterile liquid (i.e., saline, vitamin B12, lidocaine, Adequan®, etc.) into an acupuncture point. This causes constant stimulation of the acupuncture point for an extended period (until the liquid is absorbed) and it has the added benefit of the medicinal properties of the liquid used.
- Moxibustion is the use of a Chinese herb called *Artemisia vulgaris* that is rolled into a cigar shape and burned just above the acupuncture point without touching the skin. This is a warming technique that is therapeutic for older patients with chronic pain.
- Gold implantation is the injection of sterile pieces of gold, whether in a bead or wire form, into acupuncture points for permanent

Figure 23.2 Calming point needle placed on JimmyCat's head.





Figure 23.3 JimmyCat models electroacupuncture needles and machine.



Figure 23.4 JimmyCat shows acupuncture needles attached to the electrostimulator machine.

- implantation. It provides long-term stimulation for chronic conditions.
- Pneumo-acupuncture is the injection of air under the skin in the subcutaneous space to produce pressure by the air and stimulate the acupuncture points, nerves and muscles. It is used solely for muscle atrophy.
- Hemo-acupuncture is the insertion of a hypodermic needle into an acupuncture point that is located on a blood vessel to draw blood. It releases heat, toxins, and fever and can be applied like leech therapy.
- Laser acupuncture is the use of laser therapy to emit light to penetrate the tissues

and stimulate acupuncture points. It stimulates ATP production, cellular proliferation, collagen synthesis, fibroblast activity; decreases pain; and improves circulation and wound healing.

Clinical Applications Associated with Rehabilitation – Dog and Cat

Pain

Because it causes the release of endogenous opioids, acupuncture is used extensively for pain management. Acupuncture can be used as a sole agent but is more commonly practiced in conjunction with other modes of pain control as part of a multimodal protocol. In rehabilitation cases, acupuncture is commonly used after therapeutic exercises or hydrotherapy. For example, a patient that is undergoing rehabilitation for a femoral head ostectomy might start with laser therapy along the hips and lumbosacral junction, then exercises to encourage weight bearing and improve hip extension, followed by underwater treadmill, and finally acupuncture to alleviate pain. Acupuncture is reportedly effective for the treatment of various painful conditions including cervical, thoracolumbar and lumbosacral hyperpathia, chronic lameness, and degenerative joint disease (Xie and Ortiz-Umpierre (Medina), 2006).

Musculoskeletal Disorders

The most common musculoskeletal disorders treated with acupuncture are osteoarthritis, cranial cruciate ligament rupture, hip and elbow dysplasia, tendinopathies, and fractures. Therapy can be instituted before or after surgery and in many cases is used as part of the medical management protocol.

Neurologic Disorders

Acupuncture is used frequently for neurologic conditions, especially electro-acupuncture to stimulate nerve function. Common

applications include intervertebral disc disease, degenerative myelopathy, fibrocartilagenous embolism, polyneuropathy, cervical spondylomyelopathy, and degenerative lumbosacral stenosis.

Clinical Applications Associated with Rehabilitation – Equine

Pain

In horses, acupuncture is commonly used for painful conditions such as osteoarthritis, laminitis, back pain, tendinopathies, and colic. Performance horses commonly experience soft tissue pain due to exertion and/or overuse which can be successfully treated with 1–2 acupuncture treatments.

Musculoskeletal Disorders

Acupuncture is commonly sought out by performance horse owners and trainers as it provides pain relief without using pharmaceuticals. Musculoskeletal conditions that are effectively treated with acupuncture include osteoarthritis, laminitis, navicular syndrome, tendinopathies, and back pain. A study showed that electro-acupuncture was effective in alleviating chronic thoracolumbar pain in horses (Xie *et al.*, 2005).

Neurologic Disorders

Acupuncture can be used to treat neurologic conditions such as cervical spondylomyelopathy, laryngeal hemiplegia, and suprascapular neuropathy.

Safety

Although acupuncture is typically considered to be a safe and minimally invasive modality, there are some conditions that warrant caution or contraindication. For example, caution must be used when treating

a weak, debilitated, or obtunded patient. Generally, fewer needles and less stimulation are used. It is contraindicated to needle directly into skin lesions, ulcers, scar tissue, umbilicus, tumors, or masses. Specific acupuncture points around the abdomen and lumbar area are contraindicated in pregnancy. Length of needles needs to be considered, particularly in very small patients. Electro-acupuncture wire leads should not be connected through or across a known or suspected tumor or mass. Electro-acupuncture wire leads should not be connected around or across the chest of animals with pacemakers. Electro-acupuncture should not be used in patients with a history of seizures (Xie, 2007).

Role of the Veterinary Rehabilitation Technician

The role of the veterinary rehabilitation technician in TCVM includes:

- client education
- tui-na massage
- acupuncture
- patient prep
- needle prep
- needle insertion – veterinarian only
- needle manipulation/electro-acupuncture
- withdrawal of needles.

Research

The earliest scientific studies done on acupuncture focused on its analgesic effects. In the late 1970s researchers discovered that acupuncture stimulation led to an increased concentration of endogenous opioids in the serum and cerebral spinal fluid (Pan *et al.*, 1984; He, 1987). Other studies showed that naloxone, an opioid antagonist, blocked the effects of acupuncture and decreased the pain threshold in acupuncture subjects (Mayer *et al.*, 1977). These were the first studies showing that endogenous opioids

played a role in the mechanism of action of acupuncture analgesia.

Groppetti *et al.* (2011) studied the efficacy of electro-acupuncture compared with butorphanol for postoperative pain management in dogs undergoing elective ovariohysterectomy. Twelve dogs were randomly allocated into two groups. Dogs received either electro-acupuncture (16 and 43 Hz) at BL-23, BL-25, ST-36, GB-34, LI-4, LU-9, and GV-20; or butorphanol. Intra-operative cardiovascular and respiratory parameters were recorded for both groups. Plasma β -endorphin concentrations were evaluated before surgery (baseline) and up to 24 hours later. For each dog, pain was measured on a subjective pain scoring system. Plasma β -endorphin levels in dogs receiving electro-acupuncture increased significantly against baseline values after 1 and 3 hours after surgery. Moreover, the end-tidal isoflurane concentration needed for second ovary traction was significantly lower in electro-acupuncture dogs than control dogs. The dogs in the electro-acupuncture group experienced prolonged analgesia, over 24 hours at least, while 4 out of 6 dogs treated with butorphanol needed post-surgical ketorolac and tramadol supplementation for pain management (rescue analgesia). Their results showed supportive evidence for electro-acupuncture as an alternative technique to provide post-operative analgesia in dogs.

La *et al.* (2005) investigated the effects of electro-acupuncture on nerve regeneration. The sciatic nerves (specifically at a location 5 mm above the stifle joint) of 15 rabbits were crushed by a Halsted straight mosquito hemostat with 8–11 newton force for 60 seconds, and then the rabbits were divided equally into three groups. Group 1 was treated with electro-acupuncture at GB-30 and BL-40 for 25 minutes daily for 7 days. Group 2 was treated with intramuscular administration of diclofenac 15 mg daily for 7 days. Group 3 was the control group and therefore was not treated. After treatment, the distal parts of crushed nerve were examined under light microscope, the

densities of normal myelinated fibers in 0.126 mm^2 were counted, and the diameters of 20 normal myelinated fibers were measured for each rabbit. The results showed that the mean densities were 176.2 ± 5.953 in the electro-acupuncture group, 118.2 ± 10.878 in the diclofenac group, and 101.4 ± 8.548 in the control group. The mean values were significantly different between the electro-acupuncture and diclofenac groups; highly significant different between the electro-acupuncture and

control groups; and there was no significant difference between the diclofenac and control groups. There were more small myelinated fibers ($0\text{--}9 \mu\text{m}$) in the electro-acupuncture group than in the diclofenac and control groups. These results confirmed that electro-acupuncture promotes nerve regeneration whereas diclofenac does not have such an effect (La *et al.*, 2005).

Two case studies representing the use of acupuncture therapy in physical rehabilitation are described in Boxes 23.1 and 23.2.

Box 23.1 Canine case study

Jake is a 14-year-old neutered male mixed breed dog (Figure 23.5).

History

Jake was diagnosed with hip dysplasia and osteoarthritis at 6 years of age. At that time, he was started on Rimadyl® and Cosequin®. A year later, he started rehabilitation therapy and acupuncture. He also started Adequan®

injections and Duralactin® 4 years ago, as well as amantadine and Chinese herbs 1 year ago.

Physical examination

On presentation Jake was bright, alert and responsive. He was lame on the right pelvic limb, and had decreased range of motion of both hips, and pain (4/10) on extension of both hips.

Treatment plan

Recommendations consisted of acupuncture, laser therapy, manual therapy, exercises and hydrotherapy. Initially therapy was performed twice a week and currently his maintenance schedule is once a week.

Treatment

- Acupuncture (points varied at each visit but commonly used points are listed below)
 - Dry needle: Bai hui, Shen peng, Shen jiao, ST36, GB34, LIV3, LI10, GB29, BL40
 - Electro-acupuncture: BL11, BL18, BL23, GB21, Shen shu, Jian jiao + GB30, BL54 + KID1, @ 20 Hz × 20 minutes
- Laser therapy:
 - 292 Hz, 4 J/cm^2 , continuous wave, 7 minutes 12 seconds, 229.624 J total
 - Back, hips, stifles, shoulders and carpi



Figure 23.5 Jake from the canine case study undergoes acupuncture and electroacupuncture treatment.

- Manual therapy
 - Passive range of motion, full body massage, circles to hips, joint compressions to joints of pelvic limbs, standing compression of pelvic limb joints, and scapular glides
- Exercises
 - Rhythmic stabilization, cookie stretches, three-leg standing, sit to stands, backward walking, and side stepping
- Hydrotherapy

- Underwater treadmill at 0.8–1 mph (1.3–1.6 km/h) in 18 inches (46 cm) of water (level with the distal aspect of the greater trochanter) for 10–15 minutes or to tolerance

Outcome

Jake has been receiving multimodal therapy for his osteoarthritis for the past few years. This therapy has allowed him to have his pain level well-controlled and enjoy a good quality of life despite his debilitating osteoarthritis.

Box 23.2 Feline case study

Cricket is a 13-year-old spayed female domestic shorthair (Figures 23.6–23.8).

History

Cricket has been overweight most of her life and has been eating Hills W/D for the past 10 years. Over the past 6 months she has had decreased mobility and has been unwilling to get on furniture, play with other cats, and go up stairs. Also she growls at her owners when

they try petting her back and hips. She had been taking Cosequin® for the past 3 months.

Physical examination

On presentation, Cricket was bright, alert and responsive. She had a body condition score of 9/9 and weighed 22 pounds (10 kg). She exhibited pain (5/10) on range of motion of her elbows, hips, thoracolumbar spine, and lumbosacral junction.



Figure 23.6 Cricket from the feline case study receives acupuncture treatment for obesity and pain from osteoarthritis.

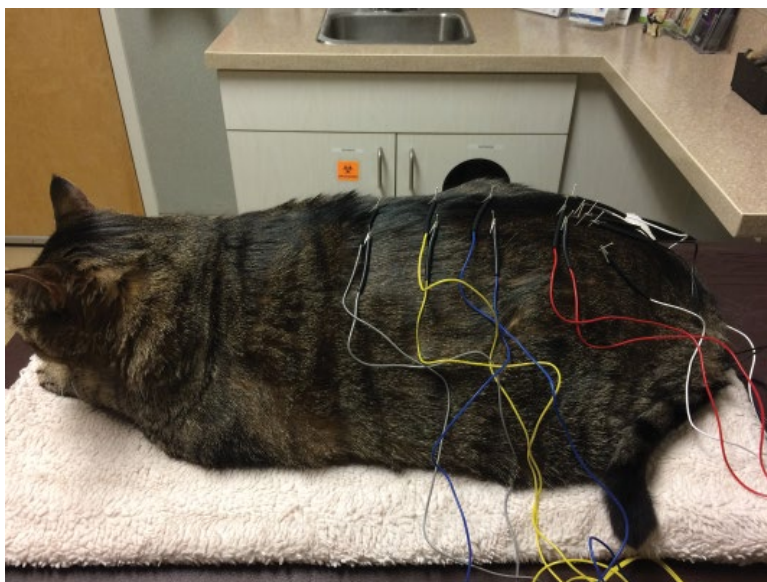


Figure 23.7 Cricket receives electroacupuncture treatment.



Figure 23.8 Cricket resting and dozing during acupuncture treatment.

Treatment plan

Recommendations consisted of continuing Cosequin®, starting gabapentin at 10 mg/kg orally every 8 hours, weekly acupuncture for 4 weeks, and switching to Hills Metabolic to promote a 10 pound (4.5 kg) weight loss.

- Initial acupuncture treatment:
 - Dry needle: Bai hui
 - Electro-acupuncture: BL18, BL20, BL23, Shen jiao, BL54 + Jian jiao @ 20Hz×20 minutes
- Follow-up acupuncture treatment (1 week after initial presentation):
 - Dry needle: Bai hui, Shen shu, Shen peng, Shen jiao
 - Electro-acupuncture: BL54 + GB30, BL18, BL20, BL23, BL25 @ 20Hz×20 minutes
- Update: Cricket is allowing her owners to pet her and jumped on their bed a few times in the past week. She is also engaging more and playing with toys
- Follow-up acupuncture treatment (2 weeks after initial presentation):

- Dry needle: Bai hui
- Electro-acupuncture: BL18, BL20, BL21, BL23, BL25, Shen shu @ 20Hz×20 minutes
- Update: Cricket is engaging more with the other cats and is also jumping up on the furniture more so than the previous week
- Follow-up acupuncture treatment: (3 weeks after initial presentation):
 - Dry needle: Bai hui, Shen shu
 - Electro-acupuncture: Jian jiao, Shen peng, Shen jiao, BL18, BL20, BL23, BL24 @ 20Hz×20 minutes

- Update: Cricket is much more active than the past 3 weeks and is comfortable on palpation

Outcome

Cricket has returned to doing the activities she used to enjoy doing and allows her owners to pet her without showing signs of pain. Her owners are continuing to work on weight loss and take her for acupuncture on an as-needed basis.

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Adjunctive Therapies Part 2: Veterinary Chiropractic

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Introduction

Lower back pain is the leading cause of disability among American workers (Meeker and Micozzi, 2001). A study published in the *British Medical Journal* concluded that chiropractic adjustment outperformed hospital outpatient management for lower back pain (Meade *et al.*, 1995). Pain is the leading reason human patients seek care from chiropractic practitioners (Leach, 1994a). Pain and pain relief are central to the discipline of chiropractic (Bove and Swenson, 2001).

It makes sense to consider applying chiropractic techniques for the relief of pain in animals within the context of physical rehabilitation. Chiropractic also aims to

restore normal joint range of motion. Adding incentive to the exploration and application of chiropractic care to animals are the challenges associated with assessing pain in non-verbal species, and the limited pharmacologic options available for managing pain. In order to better appreciate the role of the veterinary technician in the discipline of veterinary chiropractic care, it is important to have an overview understanding of chiropractic principles and practices. Although chiropractic adjustments may only be performed by a veterinarian or a chiropractor trained in animal chiropractic, the veterinary technician plays a critical role in restraint and positioning of the animal patient receiving chiropractic care.

A Brief History of Chiropractic Medicine

Chiropractic is a medical discipline based on spinal manipulation. The term chiropractic is derived from the Greek words *cheir* (“hand”) and *praxis* (“practice”). Manipulation of the body and its tissues is ancient and universal. The traditional paradigm of chiropractic reflects the following core beliefs:

- The body is self-regulating and self-healing.
- The nervous system is the master system of the body.
- Alterations in spinal movement adversely affect the nervous system’s ability to regulate function.
- Correcting, managing, or minimizing the vertebral subluxation complex via chiropractic adjustment optimizes patient health (Cleveland *et al.*, 2001).

The adaptation of chiropractic techniques for use in veterinary medicine is relatively recent (Pascoe, 2002). The American Veterinary Medical Association has published Guidelines for Complementary and Alternative Veterinary Medicine, which contains the following terminology/description about chiropractic care: “veterinary manual or manipulative therapy (similar to osteopathy, chiropractic, or physical medicine and therapy)” (AVMA, 2001).

This places chiropractic into the category of complementary care, to be used (literally) to “complement” traditional allopathic veterinary medicine and patient care. Because of its designation as a complementary medical technique, it is best to acquire informed consent from the client before beginning chiropractic treatment of an animal.

The Role of Chiropractic in Physical Rehabilitation

Chiropractic can serve as a valuable “tool in the toolchest” of the rehabilitation therapist working with animal patients. Chiropractic

provides a means to maintain spinal mobility, mobility in joints of the extremities, and function in healthy athletic and working animals. In animals experiencing clinical problems such as pain and lameness, chiropractic addresses spinal dysfunction, whether that dysfunction is the result of a primary pathology such as intervertebral disc disease (IVDD), or secondary to a lameness that leads to disrupted spinal biomechanics. The positive outcomes of chiropractic adjustment in animals undergoing physical rehabilitation include:

- restoration of proper spinal segmental function,
- restoration of proper bony relationships,
- resetting neural receptors back to healthy firing frequency,
- improving overall mobility,
- assisting neurologic healing,
- rebalancing proper muscle tone and function, and
- reducing pain (Jurek, 2013).

The Vertebral Subluxation Complex

Pathology of the spine leading to nervous system dysfunction is described as a vertebral subluxation complex (VSC). Unlike an orthopedic luxation, where bones normally engaged in a structural relationship become disrupted or distracted (e.g., coxofemoral luxation), the VSC is a term that describes a *functional* rather than a *structural* abnormality. The VSC describes a spinal segment that is restricted from moving throughout its normal range of motion. The complex describes this loss of movement within the context of one vertebra in relation to neighboring vertebrae. Gatterman (1995) specifically defines spinal subluxation as “a motion segment in which alignment, movement integrity, and/or physical function are altered though contact between the joint surfaces remains intact”. This is a reflection of neuromuscular dysfunction. Vertebrae that do not function

properly within the spinal framework generate mechanical stress, thus accelerating the wear and tear on the surrounding spinal muscles, ligaments, discs, joints, and other spinal tissues. If left untreated, pain, inflammation, tenderness to palpation, decreased mobility, and muscle spasm/tension eventually occur. After just 2 weeks of immobilization in rats, degeneration of facet joint cartilage occurred in the lumbosacral joints (Yoshida, 1989).

Detailed information about chiropractic care can be found in: Bergmann TF and Peterson DH (2011) *Chiropractic Technique: Principles and Procedures*, 3rd edn. Mosby, St. Louis, MO.

The key to appreciating the concept of the VSC and its relationship to pain is to understand the intimate relationship between structure and function. Various models of the VSC have been developed in human patients. Biped spinal biomechanics are quite different from those of quadrupeds, but some structural and functional analogies exist between biped and quadruped skeletons, allowing for the application of chiropractic principles to four-legged patients once anatomical and motion differences are accounted for.

In order to know what to treat, the practitioner must determine which spinal segment(s) are dysfunctional and in which direction(s) motion is restricted. The chiropractic examination will always include a motion palpation as a critical component in order to evaluate the relationships among spinal segments (or limb joints) as well as to identify those segments between which motion is compromised. The description of the VSC and its treatment is referred to as a “listing.” The listing vernacular provides information about the place on the animal’s body that will be contacted by the practitioner, the direction of reduced motion, as well as the direction in which the VSC will be adjusted or corrected. The listing vocabulary is taken from human chiropractic, specifically the Palmer Gonstead system (Scaringe and Cooperstein, 2001) and, in acknowledge-

ment of its origin, maintains use of human chiropractic terminology in spite of the differences in describing analogous locations on an animal’s body. One example is the use of “anterior” for “ventral” and “posterior” for “dorsal.” A complete explanation of the VSC listing nomenclature is beyond the scope of this chapter.

The Chiropractic Adjustment

The fundamental chiropractic interaction between the practitioner and the patient is the adjustment. The chiropractic adjustment involves a specific, small-amplitude, high-velocity, controlled thrust to restore motion through a specific vector by moving the joint surfaces to the anatomical limit of joint play (Leach, 1994b). It is the specificity of the adjustment, in both location and direction, which differentiates the chiropractic adjustment from other less-specific tissue manipulation. Sometimes during an adjustment there will be a sound, called an “audible,” caused by pressure changes in the joint (Leach, 1994b).

The chiropractic adjustment is focused on the functional spinal unit comprising two adjacent vertebrae, the joints that link them, the skeletal muscles that move the joints, and the supportive structures that span the distance between them. Between the elastic barrier and the anatomical barrier is a virtual/theoretical space referred to as the “physiological space,” and it is within this space that the chiropractic adjustment occurs. This is an extremely small space, which means the adjusting thrust is a very low-amplitude movement (Scaringe and Cooperstein, 2001).

The adjustment thrust is a fast, specific, and small movement applied to the affected spinal segment in the direction required to overcome the restricted motion, as diagnosed, and restore normal motion to the segment. The thrust motion stimulates mechanoreceptors and acts on the small muscles of the spinal segment (rotatory muscles),

causing them to relax or reduce spasm (Sung *et al.*, 2005). Predicting or feeling for the moment of greatest relaxation takes practice and involves palpating many, many patients to train the practitioner to most effectively diagnose and adjust VSCs (Options, 2008).

Evaluation of the Animal Chiropractic Patient

The chiropractic examination should occur prior to any adjustments. An evaluation of posture and gait, vertebral and extremity palpation, motion palpation, as well as an orthopedic and neurological evaluation are part of the examination. No matter the size of the animal chiropractic patient, the principles of evaluation, diagnosis, and adjustment are the same. The cornerstone of the chiropractic examination is the motion palpation. A systematic approach to palpation will reveal areas of discomfort or altered sensation along the spine and over the pelvis. A detailed description of a pain palpation in a dog is published elsewhere (Downing, 2011), but the key component for any species includes a systematic approach using pressure applied with the fleshy tissue over P3 of the first and second fingers of the palpator. If back pain is identified, or if areas along the back feel tense (or less pliant) than surrounding areas, the patient may be a good candidate for chiropractic evaluation and adjustment.

As the veterinarian locates the segments that need adjustment, the animal may move away from being palpated or react negatively in some way. The soft tissues in the surrounding area may spasm under the veterinarian's fingers. The veterinarian needs to begin with light pressure and slowly increase to motion palpate as needed. The veterinary technician needs to anticipate potential discomfort while restraining the patient appropriately.

The chiropractic examination should be systematic and consistent. A habitually systematic evaluation also reduces interpatient variability for the practitioner, as there is

subjectivity involved in any physical medicine technique. Evaluation and adjustments should take place in an area large enough to accommodate the patient. A non-skid surface is necessary in order for the patient to be comfortable standing. It is also important to evaluate and adjust the patient in an area free of noise or distractions. Be aware of behavioral cues that can be interpreted by the animal as signals of aggression. In keeping with the AVMA's current recommendations, clients should not restrain their animals for the chiropractic practitioner. It is the practitioner's responsibility to keep the client safe during the chiropractic diagnosis and treatment. The animal should be monitored carefully for signs of anxiety, agitation, or aggression. It will primarily be the technician's job to keep the practitioner and client safe. If the animal cannot be reassured, the chiropractic session should be stopped until it can be resumed safely and comfortably for all.

A painful dog or cat should only be adjusted after breaking the pain cycle pharmacologically. If the patient is painful, the motion palpation prior to the adjustment will be uncomfortable for the patient, and it may be challenging to determine specifically the location and nature of the VSC, and any adjustment risks exacerbating the patient's pain. In equine patients, there are fewer pain management options, and chiropractic adjustment itself can play an important role in pain relief.

The Chiropractic Adjustment and the Veterinary Technician

Companion animals are adjusted using a one-handed technique, allowing the practitioner to use the opposite hand and arm to help stabilize the patient. Appropriate stabilization allows the practitioner to isolate the affected spinal segment for adjustment, as well as to more easily take the segment to tension. In addition, appropriate stabilization of the patient ensures that most of the force of the